RVS UNIVERSITY

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Estd. under Andhra Pradesh Private Universities (Establishment and Regulation) Act, 2016

NAAC A+ GRADE ACCREDITED UNIVERSITY

R V S Nagar, Chittoor – 517 127 (AP)

RVSU - B. TECH ACADEMIC REGULATIONS (R26)

COURSE STRUCTURE AND DETAILED SYLLABI

MECHANICAL ENGINEERING

B. Tech Regular (Full - Time) Four-Year Degree Program

(For the batches admitted from the Academic Year 2026-27)

and

B.Tech. (Lateral Entry Scheme)

(For the batches admitted from the Academic Year 2027-28)

DEPARTMENT OF MECHANICAL ENGINEERING

SCHOOL OF ENGINEERING AND TECHNOLOGY (SOET)

Website: www.rvsu.org **E-mail**: vicechancellor@rvsu.org

FOREWORD

The Academic Regulations and Syllabi of RVS University, are founded on a commitment to academic excellence, ethical values, innovation and holistic development. Our academic philosophy places the learner at the centre of the educational process, fostering knowledge, professional competence, intellectual curiosity, creativity, character and social responsibility.

The University is firmly committed to Outcome-Based Education (OBE), with clearly articulated learning outcomes and measurable competencies that equip graduates to address the evolving challenges of society, industry and the global knowledge economy.

The curricula integrate disciplinary knowledge with critical thinking, experiential and interdisciplinary learning, research, emerging technologies, entrepreneurship and real-world problem solving. The University seeks to promote academic freedom with accountability, encouraging intellectual inquiry while upholding academic standards and institutional responsibility.

These Academic Regulations establish a transparent and rigorous framework for curriculum design, teaching-learning, assessment, academic progression and governance. The syllabi are conceived as dynamic academic instruments, subject to continuous review and enrichment in response to advances in knowledge, professional requirements, societal needs, national priorities and global developments.

We believe that education must extend beyond employability to develop capable, ethical, innovative and socially responsible professionals and citizens. I am confident that these Academic Regulations and Syllabi will provide a strong foundation for cultivating a culture of quality, academic excellence, innovation, continuous improvement and meaningful contribution to nation-building.

Dr. Ravuri Venkataswamy
Chancellor

VISION, MISSION AND OBJECTIVES

VISION:

To be recognized as world class University that promotes advanced research, contemporary teaching, experiential learning and extension.

MISSION:

- Advance student knowledge through research integrated with education to benefit and address the diversity.
- Undertake challenging problems in multi-disciplinary environments through developing students in domain and soft skills.
- Imbibe a culture with human values and professionalism to develop as creative resources for the nation.
- Establish an ecosystem for faculty and staff for continuous quality improvement.
- Collaborate with industry and premiere educational institutions for quality research and innovation.

OBJECTIVES:

- Establish Centres of excellence in teaching, training and learning.
- Create facilities for quality research which contributes for institutional and national development.
- Developing an ecosystem for innovation and entrepreneurship.
- Collaborate with non-government agencies and industry to undertake extension and social responsibility.

CORE VALUES:

- **Excellence:** Commitment to continuous improvement and innovation in Teaching, Learning, research and extension.
- **Ethics:** Pledge to honesty, integrity, mutual respect, transparency and accountability.
- **Diversity:** Respect to all aspects of diversity.
- **Student First:** Student centric planning and development.
- **Collaboration:** Stakeholder partnership for holistic institutional development and synergistic growth.

RVSU - B. Tech Academic Regulations (R26) (Regular-Fulltime)

(Effective for the students admitted into I year from the Academic Year **2026-27** onwards)

RVS University offers Regular full-time (Four-Year) Bachelor of Technology (B. Tech.) Degree programme under Choice Based Credit System (CBCS) in various branches of Engineering and Technology.

The RVS University shall confer B. Tech Undergraduate degree to candidates who are admitted to the Bachelor of Technology Program and fulfill all the requirements for the award of the degree.

1. **Applicability:**
   All the rules specified herein, approved by the Academic Council, shall be in the force and applicable to the students admitted from the Academic Year 2026-2027 onwards. Any reference to "University" in these rules and regulations stands for RVS University.
2. **Extent:**
   All the rules and regulations, specified hereinafter shall be read as a whole for the purpose of interpretation. As and when a doubt arises, the interpretation of the Vice Chancellor shall be final and ratified by the Academic Council in the forthcoming meeting. As per the requirements of statutory bodies, Vice Chancellor shall be the Chairman, Academic Council.
3. **Award of the Degree**
   a) **Award of the B.Tech. Degree / B.Tech. Degree with a Minor if he/she fulfils the following:**
      i. Pursues a course of study for not less than four academic years and not more than eight academic years. However, for the students availing Gap year facility this period shall be extended by two years at the most and these two years would in addition to the maximum period permitted for graduation (Eight years).
      ii. Registers for 169 and secures all 169 credits

   b) **Award of B.Tech. degree with Honors if he/she fulfils the following:**
      i. Student secures additional 18 credits fulfilling all the requisites of a B.Tech. program i.e., 169 credits.
      ii. Registering for Honors is optional.
      iii. Honors is to be completed simultaneously with B.Tech. programme.

   c) **Award of B.Tech. degree with Minors**
      i. Student secures additional 18 credits fulfilling all the requisites of a B.Tech. program i.e., 169 credits.
      ii. Registering for Minors is optional.
      iii. Minors is to be completed simultaneously with B.Tech. programme
4. Students, who fail to fulfil all the academic requirements for the award of the degree within eight academic years from the year of their admission, shall forfeit their seat in B.Tech. course and their admission stands cancelled. This clause shall be read along with clause 3 a) i).
5. **Admissions**
   Admission to the B. Tech Program shall be made subject to the eligibility, qualifications and specialization prescribed by the A.P. State Government / RVS University from time to time. Admissions shall be made either based on the merit rank obtained by the student in the common entrance examination conducted by the A.P. Government and / or RVS University or any other order of merit approved by the A.P. Government, subject to reservations as prescribed by the Government / RVS University from time to time.
6. **Program related terms**
   Credit: A unit by which the course work is measured. It determines the number of hours of instruction required per week. One credit is equivalent to one hour of teaching (Lecture / Tutorial) or two hours of practical work / field work per week.

Credit Definition:

1Hr. Lecture(L) per week	1 Credit
1Hr.Tutorial (T) per week	1 Credit
1Hr.Practical (P) per week	0.5 Credit
2Hrs.Practical (Lab) per week	1 Credit

a) ***Academic Year:*** Two consecutive (one odd + one even) semesters constitute one academic year.

b) ***Choice Based Credit System (CBCS):*** The CBCS provides a choice for students to select from the prescribed courses.

7. Semester/Credits:

i) A semester comprises of 15 weeks and an academic year is divided into two semesters.

ii) The summer term is for eight weeks during summer vacation. Internship / apprenticeship / work-based vocational education and training can be carried out during the summer term, especially by students who wish to exit after two semesters or four semesters of study.

iii) Regular courses may also be completed well in advance through MOOCs satisfying prerequisites.

8. Structure of the Undergraduate Programme:

All courses offered for the undergraduate program (B.Tech.) are broadly classified as follows:

S. No.	Category	Break-up of Credits (Total 169)	Percentage of total Credits	AICTE Recommendation (%)
1.	Humanities and Social Science including Management (HS)	15	9%	8 – 9%
2.	Basic Sciences (BS)	22	13 %	12 -16%
3.	Engineering Sciences (ES)	24	15%	10– 18%
4.	Professional Core (PC)	57	33 %	30– 36%
5.	Electives – Professional (PE) & Open (OE); Domain Specific Skill Enhancement Courses (SEC)	35	20 %	19 -23%
6.	Internships & Project work (PR)	16	10 %	8 –11%
7.	Mandatory Courses (MC)	Non-credit	Non-credit	-

The total degree credits become 169, including 2 credits for online Examination. The syllabus for this online examination shall comprise core courses studied by the student up to IV B. Tech, I Semester and this examination will be conducted along with IV B. Tech, I semester Regular Examinations.

9. Course Classification:

All subjects / courses offered for the undergraduate programme in Engineering & Technology (B.Tech. degree programmes) are broadly classified as follows:

S. No.	Broad Course Classification	Course Category	Description
1.	Foundation Courses	Foundation courses	Includes Mathematics, Physics and Chemistry; fundamental engineering courses; humanities, social sciences and management courses
2.	Core Courses	Professional Core Courses (PC)	Includes subjects related to the parent discipline / department / branch of

			Engineering
3.	Elective Courses	Professional Elective Courses (PE)	Includes elective subjects related to the parent discipline / department / branch of Engineering
		Open Elective Courses (OE)	Elective subjects which include interdisciplinary subjects or subjects in an area outside the parent discipline / department / branch of Engineering
		Domain specific skill enhancement courses (SEC)	Interdisciplinary / job-oriented / domain courses which are relevant to the industry
4.	Project & Internships	Project	B.Tech. Project or Major Project
		Internships	Summer Internships–Community based and Industry Internships; Industry oriented Full Semester Internship
5.	Mandatory Courses	Mandatory non-credit courses	Covering subjects of developing desired attitude among the learners
6.	Audit Courses	Non-Mandatory non-credit courses	Covering subjects which develop additional skills and values among the learners

10. Programme Pattern

i. Total duration of B. Tech (Regular) Programme is four academic years.
ii. Each academic year of study is divided into two semesters.
iii. Minimum number of instructional weeks in each semester is 15 weeks.
iv. There shall be mandatory student induction program for fresher's, with a three-week duration before the commencement of first semester. Physical activity, Creative Arts, Universal Human Values, Literary, Proficiency Modules, Lectures by Eminent People, Visits to local Areas, Familiarization to Dept./Branch & Innovations etc., are included as per the guidelines issued by AICTE.
v. Health / wellness / yoga / sports / NSS / NCC / Scouts & Guides / Community service activities are made mandatory as credit courses for all the undergraduate students.
vi. Courses like Environmental Sciences, Indian Knowledge System, Cybersecurity Awareness for Engineers, Gender Sensitization, Technical Paper Writing & IPR are offered as non-credit mandatory courses for all the undergraduate students. Audit Courses are also introduced as non-Mandatory non-Credit courses.
vii. Design Thinking & Tinkering Labs are made mandatory as credit courses for all the undergraduate students.
viii. A comprehensive online examination will be conducted in IV Year – I Semester for 2 credits
ix. Increased flexibility for students through an increase in the elective component of the curriculum, with 05 Professional Elective courses and 04 Open Elective courses.
x. Professional Elective Courses, include the elective courses relevant to the chosen specialization / branch. Proper choice of professional elective courses can lead to students specializing in emerging areas within the chosen field of study.
xi. A total of 04 Open Electives is offered in the curriculum.
xii. While choosing the electives, students shall ensure that they do not opt for the courses with syllabus contents similar to courses already pursued.
xiii. A pool of interdisciplinary / job-oriented / domain skill courses which are relevant to the industry are integrated into the curriculum of all disciplines. There shall be 05 skill-oriented courses offered during II B. Tech I Semester to IV B. Tech I Semester. Among the five skill courses, four courses shall focus on the basic and advanced skills related to the domain/interdisciplinary courses and the other shall be a soft skills course.
xiv. Students shall undergo mandatory summer internships, for a minimum of six-eight weeks duration at the end of second and third year of the programme. The internship at the end of second year shall be community oriented and industry internship at the end of third year.
xv. There shall also be mandatory full internship in the final semester of the

programme along with the Major project work.

xvi. Undergraduate degree with Honors and Minors is introduced by the University for the students having good academic record.

xvii. The Dean shall take measures to implement Virtual Labs (https://www.vlab.co.in) which provide remote access to labs in various disciplines of Engineering and will help student in learning basic and advanced concepts through remote experimentation. Student shall be made to work on virtual lab experiments during the regular labs.

xviii. The Dean shall assign a faculty advisor / mentor after admission to a group of students from same department to provide guidance in courses registration / career growth / placements / opportunities for higher studies / GATE / other competitive exams.

xix. Preferably 25% of course work for the theory courses in every semester shall be conducted in the blended mode of learning.

11. Evaluation Process

The performance of a student in each semester shall be evaluated subject wise with a maximum of 100 marks for theory and 100 marks for practical subject. Summer Internships shall be evaluated for 50 marks, Full Internship in final semester shall be evaluated for 50 marks & Project work in final semester shall be evaluated for 200 marks, mandatory courses with no credits shall be evaluated for 40 marks through mid-semester examinations.

A student has to secure not less than 35% of marks in the end examination and a minimum of 40% of marks in the sum total of the mid semester and end examination marks taken together for the theory, practical, design, drawing subject or project etc. In case of a mandatory course, he / she should secure 40% of the total marks.

Theory Courses

Assessment Method	Marks
Continuous Internal Assessment	40
Semester End Examination	60
Total	**100**

i) For theory subject, the distribution shall be 40 marks for Internal Evaluation and 60 marks for the End-Examination.

ii) For practical subject, the distribution shall be 40 marks for Internal Evaluation and 60 marks for the End-Examination.

iii) If any course contains two different branch subjects, the syllabus shall be written in two parts with 3 units each (Part-A and Part-B) and external examination question paper shall be set with two parts each for 30 marks.

iv) If any subject is having both theory and practical components, they will be evaluated separately as theory subject and practical subject. However, the marks scored in both theory and practical components shall be added and the same may be condensed to 100 marks for calculation of grade point.

a) Continuous Internal Evaluation

i) For theory subjects, during the semester, there shall be two mid-term examinations. Each mid-term examination shall be evaluated for 40 marks of which 10 marks for objective paper (20 minutes duration), 25 marks for subjective paper (90 minutes duration) and 5 marks for assignment.

ii) Objective paper shall contain for 05 short answer questions with 1 mark each or maximum of 10 multiple choice questions for 05 marks. Subjective paper shall contain 3 either or type questions (totally six questions from 1 to 6) of which student has to answer one from each either-or type of questions. Each question carries 10 marks. The total marks for the subjective paper will be 30 marks.

Note:

- The objective paper shall be prepared in line with the quality of competitive examinations questions.
- The subjective paper shall contain 3 either or type questions of equal

weightage of 10 marks. Any fraction shall be rounded off to the next higher mark.

- The objective paper examination shall be conducted along with the subjective paper examination.
- Assignments shall be in the form of problems, mini projects, design problems, slip tests, case studies, quizzes or any other relevant and suitable activity prescribed by the department., depending on the course content. The first assignment for the first two units and the second assignment for the next three units and each assignment shall be evaluated for 5 marks. The average of two assignments shall be considered for assignment marks.

iii) If the student is absent for the mid semester examination, no re-exam shall be conducted and mid semester marks for that examination shall be considered as zero.

iv) First midterm examination shall be conducted for I, II units of syllabus with one either or type question from each unit and third either or type question from both the units. The second midterm examination shall be conducted for III, IV and V units with one either or type question from each unit.

v) Final mid semester marks shall be arrived at by considering the marks secured by the student in both the mid examinations with 80% weightage given to the better mid exam and 20% to the other.

For Example:

Marks obtained in first mid: 25 Marks
obtained in second mid: 20
Final mid semester Marks: (25x0.8) + (20x0.2) =24

If the student is absent for any one midterm examination, the final mid semester marks shall be arrived at by considering 80% weight age to the marks secured by the student in the appeared examination and zero to the other. For Example:

Marks obtained in first mid: Absent Marks
obtained in second mid:25
Final mid semester Marks: (25x0.8) +(0x0.2) =20

b) End Examination Evaluation:

End examination of theory subjects shall have the following pattern:

i) There shall be 6 questions and all questions are compulsory.

ii) Question I shall contain 5 compulsory short answer questions for a total of 10 marks such that each question carries 2 marks. There shall be 1 short answer question from each unit.

iii) The questions from 2 to 6 shall be set by covering one unit of the syllabus for each question. In each of the questions from 2 to 6, there shall be either / or type questions of 10 marks each. Student shall answer any one from each either-or type questions.

End examination of theory subjects consisting of two parts of different subjects, for Example: Basic Electrical & Electronics Engineering shall have the following pattern:

i) Question paper shall be in two parts viz., Part-A and Part-B with equal weightage of 30 marks each.

ii) In each part, question 1 shall contain 6 compulsory short answer questions for a total of 6 marks such that each question carries 1mark.

iii) In each part, questions from 2 to 4, there shall be either / or type questions of 8 marks each. Student shall answer any one from each either-or type questions.

iv) The questions from 2 to 4 shall be set by covering one unit of the syllabus for each question.

c) For practical courses, there shall be a continuous evaluation during the semester for 40 sessional marks and end examination shall be for 60 marks.

i) Day-to-day work in the laboratory shall be evaluated for 20 marks by the concerned laboratory teacher based on the record / viva and 20 marks for the internal test.

ii) The end examination shall be evaluated for 60 marks, conducted by the concerned laboratory teacher and a senior faculty member in the subject from the same department.

a. Procedure: 20marks
b. Experimental work & Results: 30 marks
c. Viva-voce: 10marks.

In a practical subject consisting of two parts (Eg: Basic Electrical & Electronics Engineering Lab), the end examination shall be conducted for 60 marks as a single laboratory in 3 hours. Mid semester examination shall be evaluated as above for 40 marks in each part and final mid semester marks shall be arrived by considering the average of marks obtained in two parts.

iii) For the subject having design and/or drawing, such as Engineering Drawing, the distribution of marks shall be 40 for mid semester evaluation and 60 for end examination.

Assessment Method	Marks
Continuous Internal Assessment	40
Semester End Examination	60
Total	**100**

Day-to-day work shall be evaluated for 10 marks by the concerned subject teacher based on the reports/submissions prepared in the class. And there shall be two mid-term examinations in a semester for duration of 2 hours each for 30 marks with weightage of 80% to better mid marks and 20% for the other. The subjective paper shall contain 3 either or type questions of equal weightage of 10 marks. There shall be no objective paper in mid semester examination. The sum of day-to-day evaluation and the mid semester marks will be the final sessional marks for the subject.

The end examination pattern for Engineering Graphics, shall consists of 5 questions, either/or type, of 12 marks each. There shall be no objective type questions in the end examination. However, the end examination pattern for other subjects related to design/drawing, multiple branches, etc is mentioned along with the syllabus.

iv) There shall be no external examination for mandatory courses with zero credits. However, attendance shall be considered while calculating aggregate attendance and student shall be declared to have passed the mandatory course only when he/she secures 40% or more in the internal examinations. In case, the student fails, a re-examination shall be conducted for failed candidates for 40 marks satisfying the conditions mentioned in item 1 & 2 of the regulations.

v) The laboratory records and mid semester test papers shall be preserved for a minimum of 3 years in the concerned Department as per the University norms and shall be produced to the Committees of the University as and when the same are asked for.

Note:

- The end-semester examinations shall be conducted based on the theoretical components of the prescribed syllabus.
- Assignments shall be designed to incorporate AI-integrated elements of the syllabus to promote applied and experiential learning.
- The mid-semester examinations may include a combination of both theoretical content and AI-integrated components

d) For theory and practical combined courses, the theory and practical components shall be evaluated for 100 marks each. The evaluation for theory component shall be conducted in line with theory courses. The evaluation for practical component shall be conducted in line with practical courses. However, to declare pass in the course, the student has to satisfy minimum academic requirements (pass marks) in both theory and laboratory components as stipulated for theory and laboratory courses. The marks scored in theory component and practical component shall be interpolated to 50 marks each separately and these interpolated marks in theory component (50 marks) and practical component (50 marks) shall be combined for awarding of grade in the concerned course.

12. Skill oriented Courses

i) There shall be five skill- oriented courses offered during II B. Tech I semester to IV B. Tech I semester.

ii) Out of the five skill courses two shall be skill-oriented courses from the same domain. Of the remaining three skill courses, one shall be a soft skill course and the remaining two shall be skill-advanced courses from the same domain / Interdisciplinary / Job oriented.

iii) The course shall carry100 marks and shall be evaluated through continuous assessments during the semester for 40 sessional marks and end examination shall be for 60 marks. Day-to-day work in the class / laboratory shall be evaluated for 40 marks by the concerned teacher based on the regularity / assignments / viva/ mid semester test. The end examination similar to practical examination pattern shall be conducted by the concerned teacher and an expert in the subject nominated by the Dean.

iv) The Head of the Department shall identify a faculty member as coordinator for the course. A committee consisting of the Head of the Department, coordinator and a senior Faculty member nominated by the Head of the Department shall monitor the evaluation process. The marks/grades shall be assigned to the students by the above committee based on their performance.

v) The student shall be given an option to choose either the skill courses being offered by the School or to choose a certificate course being offered by industries/Professional bodies or any other accredited bodies. If a student chooses to take a Certificate Course offered by external agencies, the credits shall be awarded to the student upon producing the Course Completion Certificate from the agency. A committee shall be formed at the level of the School to evaluate the grades/marks given for a course by external agencies and convert to the equivalent marks/grades.

vi) The recommended courses offered by external agencies, conversions and appropriate grades/marks are to be approved by the University at the beginning of the semester. The Dean of the School shall forward such proposals to the University for approval.

vii) If a student prefers to take a certificate course offered by external agency, the department shall mark attendance of the student for the remaining courses in that semester excluding the skill course in all the calculations of mandatory attendance requirements upon producing a valid certificate as approved by the University.

13. Massive Open Online Courses (MOOCs):

To promote self-learning and lifelong learning, students are encouraged to enroll in Massive Open Online Courses (MOOCs) offered through **SWAYAM, SWAYAM Plus, and NPTEL** or other University-approved platforms.

A student may earn up to 40% of the total credits prescribed in a semester through MOOCs, subject to the University's guidelines and equivalence requirements.

Further, every student shall mandatorily complete atleast one MOOC course before the commencement of the IV Year II Semester. In addition, the Fifth Professional Elective (PE-V) and the Fourth Open Elective (OE-IV) shall be mandatorily completed through the SWAYAM/SWAYAM Plus platforms. This provision is intended to facilitate students in undertaking the Long-Term Internship during the final semester. Accordingly, every student shall mandatorily complete a total of three MOOC courses for the award of the B.Tech. degree. The following are guidelines for MOOC courses.

i. Students may earn a maximum of 40% of the credits in a semester through MOOCs for PE, OE, and SEC. Further, students of III and IV year are also permitted to pursue Core Courses (CC) through MOOCs.
ii. All the courses for Honors and Minors may be studied through MOOCs platform.
iii. Students shall register for MOOCs courses of a minimum duration of either 8 weeks or 12 weeks with the prior approval of the Head of the Department. Credits shall be awarded as follows:
   a) 2 credits for an 8-week course
   b) 3 credits for a 12-week course
iv. The NCrF rating for the Course is 4.5 for UG program
v. The Head of the Department shall nominate a faculty mentor to monitor the academic progress of students enrolled in MOOCs. It is also responsibility of the

mentor to see that the registered subjects are in the approved list provided by the university.

vi. Students shall have the option to choose either the regular offline (classroom) mode or the SWAYAM mode for a particular subject. However, a student shall not be permitted to opt for both modes simultaneously for the same subject. Students opting for SWAYAM shall be exempted from attending regular classroom sessions for that subject, and attendance shall not be counted. Further, the Departments should not insist the student to opt SWAYAM course compulsorily.

vii. All students in a class are not required to opt for the same SWAYAM course. Departments shall arrange regular classes for students opting for offline mode, while the same faculty member may act as a mentor for students enrolled in the SWAYAM course. If all students opt for a SWAYAM course, the subject need not be included in the timetable; however, a mentor shall be assigned to address student queries as per mutual convenience.

viii. Students who completed the prescribed assignments successfully and secured at least 40% of the internal marks shall be permitted to appear for the end-semester examination conducted either by the National Testing Agency (NTA)/NPTEL or by the University. University conducted examinations shall be held along with regular end-semester examinations, with a weightage of 60% for the end-term examination and 40% for assignments and quizzes evaluated by the SWAYAM Course Coordinator.

ix. Credits shall be awarded only upon submission of a successful course completion certificate. Internal marks obtained through SWAYAM shall be extrapolated to 40 and external marks interpolated to 60, and shall exactly correspond to the marks secured by the student in the SWAYAM course.

x. In case where University prescribed passing marks are higher than SWAYAM's minimum requirements, the University norms shall prevail. In such cases, even if a student submits a SWAYAM certificate but fails to secure the University's minimum passing marks, the student shall be declared unsuccessful and must appear for the supplementary examination.

xi. In cases where a student obtains less than the prescribed minimum internal marks (16 out of 40) in a SWAYAM course, and the said course is not offered again in the subsequent semester, the University may, through a special notification, conduct a full course supplementary examination for 100 marks during the final semester of the programme. The examination shall be conducted in accordance with the University question paper pattern for **60 marks**, strictly following the University prescribed syllabus and it will be extrapolated to 100.

xii. For SWAYAM courses where end-term examinations are conducted by the University, students shall not be issued a SWAYAM certificate.

xiii. If a SWAYAM course carries more credits than the corresponding RVSU course, credit transfer shall be restricted to the number of credits prescribed by RVSU to maintain uniformity in total credit requirements. Credit transfer shall not be permitted if the SWAYAM course carries fewer credits than the RVSU course.

Students who have qualified in the proctored examinations conducted through MOOCs platform can apply for credit transfer as specified and are exempted from appearing internal as well as external examination (for the specified equivalent credit course only) conducted by the University.

Necessary amendments in rules and regulations regarding adoption of MOOC courses would be proposed from time to time.

14. **Credit Transfer Policy**

Adoption of MOOCs is mandatory, to enable Blended model of teaching-learning as also envisaged in the NEP2020. As per University Grants Commission (Credit Frame work for Online Learning Courses through SWAYAM) Regulation, 2016, the University shall allow up to a maximum of 40% of the credits in a semester being offered in a particular programme i.e., via Swayam/Swayam Plus courses through MOOCs platform.

i) The University shall offer credit mobility for MOOCs and give the equivalent credit weightage to the students for the credits earned through online learning courses.
ii) Student registration for the MOOCs shall be only through the respective department, it is mandatory for the student to share necessary information with the department.
iii) Credit transfer policy will be applicable to the Professional & Open Elective courses only.
iv) The concerned department shall identify the courses permitted forced it transfer.
v) The University/Departments shall notify at the beginning of semester the list of the online learning courses eligible for credit transfer.
vi) The Departments shall designate a faculty member as a Mentor for each course to guide the students from registration till completion of the credit course.
vii) The University shall ensure no overlap of MOOC exams with that of the University examination schedule. In case of delay in results, the University will re-issue the marks sheet for such students.
viii) Student pursuing courses under MOOCs shall acquire the required credits only after successful completion of the course and submitting a certificate issued by the competent authority along with the percentage of marks and grades.
ix) The Departments shall submit the following to the examination section of the university:
    a) List of students who have passed MOOC courses in the current semester along with the certificate of completion.
    b) Undertaking form filled by the students for credit transfer.
x) The University shall resolve any issues that may arise in the implementation of this policy from time to time and shall review its credit transfer policy in the light of periodic changes brought by UGC, SWAYAM, NPTEL and State Government.

15. Academic Bank of Credits (ABC)

The University is implementing Academic Bank of Credits (ABC) to promote flexibility in curriculum as per NEP 2020 to

i. Provide option of mobility for learners across the universities of their choice.
ii. Provide option to gain the credits through MOOCs from approved digital platforms.
iii. Facilitate award of certificate / diploma / degree in line with the accumulated credits in ABC.
iv. Execute Multiple Entry and Exit system with credit count, credit transfer and credit acceptance from students' account.

16. Mandatory Internships

Summer Internships: Two summer internships with a minimum of 06-08 weeks duration (2 credits each), done at the end of second and third years, respectively are mandatory. It shall be completed in collaboration with local industries, Govt. Organizations, construction agencies, Power projects, software MNCs or any industries in the areas of concerned specialization of the Undergraduate (B. Tech) program. One of the two summer internships at the end of second year (Community Service Project) shall be society oriented and shall be completed in collaboration with government organizations / NGOs & others. The other internship at the end of third year is Industry Internship and shall be completed in collaboration with Industries. The student shall register for the internship as per course structure after commencement of academic year. The guidelines issued by the APSCHE / University shall be followed for carrying out and evaluation of Community Service Project and Industry Internship.

Evaluation of the summer internships shall be through the departmental committee. A student will be required to submit a summer internship report to the concerned department and appear for an oral presentation before the departmental committee comprising of Head of the Department, Mentor of the internship and a senior faculty member of the department. A certificate of successful completion from industry shall be included in the report. The report and the oral presentation shall carry 50% weightage each. It shall be evaluated for 50 external marks. There shall be no internal marks for Summer Internship. A student shall secure minimum 40% of marks for successful completion. In case, if a student fails, he/she shall reappear as and when semester supplementary examinations are conducted by the University.

Full Semester Internship and Project work:
In the final semester, the student should mandatorily register and undergo internship and in parallel he / she should work on a project with well-defined objectives. At the end of the semester the candidate shall submit an internship completion certificate and a project report. A student shall also be permitted to submit project report on the work carried out during the internship.

The Long-Term Industry Internship undertaken in the final semester (IV Year – II Semester) shall be evaluated for 50 marks by a committee consisting of the Head of the Department along with two senior faculty members of the department. Further, every student shall undertake at least one Industry Visit between the III B. Tech I Semester and IV B. Tech I Semester, as part of the mandatory internships and practical training requirements.

The project report shall be evaluated with an external examiner. The total marks for project work 200 marks (8 credits) and distribution shall be 80 marks for internal and 120 marks for external evaluation. The supervisor assesses the student for 40 marks (Report: 20 marks, Seminar: 20 marks). At the end of the semester, all projects shall be showcased at the department for the benefit of all students and staff and the same is to be evaluated by the departmental Project Review Committee consisting of supervisor, a senior faculty and HOD for 40 marks. The external evaluation of Project Work is a Viva-Voce Examination conducted in the presence of internal examiner (Supervisor), Head of the department and external examiner appointed by the University and is evaluated for 120 marks.

The Departments shall facilitate and monitor the student internship programs. Completion of internships is mandatory, if any student fails to complete internship, he/she will not be eligible for the award of degree. In such cases, the student shall repeat and complete the internship.

17. Student Development Program (SDP) – Department-Wise Workshop (Mandatory Non-Credit Course)

Each Department shall design and conduct a **Student Development Programme (SDP) / Department-Wise Workshop** relevant to its discipline for all B. Tech. students. The programme shall carry **Zero Credits** and shall be conducted **at the end of the III B. Tech. I Semester**.

The following guidelines shall govern the conduct and evaluation of the programme:

i. **Programme Planning:** The Head of the Department (HoD) shall prepare and notify the workshop schedule, syllabus/content, learning outcomes, and assessment plan at least **two weeks prior to the commencement** of the programme.
ii. **Attendance:** Attendance in all workshop sessions shall be **mandatory**. Students shall attend the programme as per the notified schedule.
iii. **Evaluation:** The programme shall be evaluated for a total of **50 marks**, comprising:
    - **25 marks** for continuous assessment based on participation, assignments, practical activities, presentations, or other evaluation components prescribed by the Department; and
    - **25 marks** for a final examination to be conducted on the **last day of the workshop**.
iv. **Pass Requirements:** A student shall be declared to have **successfully completed** the programme only if he/she secures 40% marks in both continuous assessment and final examination taken together:
v. **Submission of Marks:** The HoD shall submit the finalized marks list to the **Controller of Examinations, RVSU,** in the prescribed format within the timeline notified by the University.
vi. **Certification:** A **Workshop Completion Certificate** shall be issued only to those students who have successfully fulfilled the prescribed attendance and pass requirements.
vii. **Supplementary Examination:** A student who fails to secure the minimum prescribed marks shall be required to appear for the **supplementary examination** conducted by the University in the subsequent semester, as per the University rules

and schedule.

Successful completion of the **Student Development Programme (SDP)/Department-Wise Workshop** is **mandatory for the award of the B.Tech. degree**, and its completion status shall be reflected in the student's academic records.

18. Guidelines for offering a Minor Degree

To promote interdisciplinary knowledge among the students, the students admitted into B. Tech. in a major stream/branch are eligible to obtain degree in Minor in another stream.

i) Minor Degree is introduced in the curriculum of all B. Tech. programs offering a major degree and is applicable to all B. Tech (Regular and Lateral Entry) students admitted in Engineering & Technology.

ii) A student shall earn additional 18 credits (comprising 5 theory courses and 2 laboratory / project courses, offered in tracks) for award of B. Tech. (Minor degree). This is in addition to the credits essential for obtaining the Undergraduate (B. Tech) degree in Major Discipline (i.e., 169 credits).

iii) A student is permitted to register for Minor degree in II B. Tech II semester after the results of II B. Tech I Semester are declared and students may be allowed to take maximum two subjects per semester pertaining to the Minor Degree from II B. Tech II Semester onwards.

iv) The Dean of the School shall arrange separate class work and timetable of the courses offered under Minor Degree program.

v) Courses that are used to fulfil the student's primary major (Open Elective) may not be double counted towards the Minor Degree. Courses with content substantially equivalent to courses in the student's primary Major Degree may not be counted towards the Minor Degree.

vi) Students can complete the courses offered under Minor Degree either in the University or in online platforms like SWAYAM with a minimum duration of 12 weeks for a 3-credit course and 8 weeks duration for a 2-credit course satisfying the criteria for credit mobility. If the courses under Minor Degree are offered in conventional mode, then the teaching and evaluation procedure shall be similar to regular B. Tech courses.

vii) The attendance for the registered courses under Minor Degree and regular courses offered for Major degree in a semester are to be considered separately.

viii) A student shall maintain an attendance of 75% in all registered courses under Minor Degree to be eligible for attending semester end examinations.

ix) A student registered for Minor Degree shall pass in all subjects that constitute the requirement for the Minor degree program. No class/division (i.e., second class, first class and distinction, etc.) shall be awarded for Minor degree programme.

x) If a student drops or is terminated from the Minor degree program, the additional credits so far earned cannot be converted into open or core electives; they will remain extra. However, such students will receive a separate grade sheet mentioning the additional courses completed by them.

xi) The Minor Degree will be mentioned in the degree certificate. Every Department in the School shall offer Minor Degree program and A Minor Degree course on Quantum Computing shall be offered for all branches, with the curriculum framed by the CSE Department.

xii) A student who fails in any course of the Minor Degree Programme shall be permitted to appear for the corresponding supplementary examination(s) as per the University rules. However, the student shall successfully complete all the prescribed courses of Minor Degree program before Completion of Major Degree Programme. Failure to do so shall result in the cancellation of admission to the Minor Degree Programme, and no Minor credential shall be awarded.

xiii) All the courses offered under Minor Degree may also be pursued through MOOCs. The Bloom's Taxonomy level for Minor courses shall be 3 and below.

19. Guidelines for offering Honors Degree

The objective of introducing B. Tech. (Honors) is to facilitate the students to choose additionally the specialized courses of their choice and build their competence in a specialized area in the UG level. The programme is a best choice for academically excellent students having good academic record and interest towards higher studies and research.

i. Honors Degree is introduced in the curriculum of all B. Tech. programs offering a major degree and is applicable to all B. Tech (Regular and Lateral Entry) students admitted in Engineering & Technology.

ii. Students with CGPA of 7 and no back logs upto II B. Tech I semester are eligible to register for Honors Degree programs.

iii. A student shall earn additional 18 credits (comprising 5 theory courses and 2 laboratory / project courses, offered in tracks) for award of B. Tech. (Honors) degree from same branch / department / discipline registered for major degree. This is in addition to the credits essential for obtaining the Undergraduate degree in Major Discipline (i.e., 169 credits).

iv. A student is permitted to register for Honors Degree in II B. Tech II Semester after the results of II B. Tech I Semester are declared and students may be allowed to take maximum two subjects per semester pertaining to the Honors Degree from II B. Tech II Semester onwards.

v. The concerned Head of the Departments shall arrange separate class work and time table of the courses offered under Honors Degree program.

vi. Courses that are used to fulfil the student's primary major may not be double counted towards the Honors Degree Courses with content substantially equivalent to courses in the student's primary Major may not be counted towards the Honors Degree.

vii. Students can complete the courses offered under Honors Degree either in the University or in online platforms like SWAYAM with a minimum duration of 12 weeks for a 3-credit course and 8 weeks duration for a 2-credit course satisfying the criteria for credit mobility. If the courses under Honors Degree are offered in conventional mode, then the teaching and evaluation procedure shall be similar to regular B. Tech courses.

viii. The attendance for the registered courses under Honors Degree and regular courses offered for Major degree in a semester are to be considered separately.

ix. A student shall maintain an attendance of 75% in all registered courses under Honors Degree to be eligible for attending semester end examinations.

x. A student registered for Honors Degree shall pass in all subjects that constitute the requirement for the Honors degree program. No class/division (i.e., second class, first class and distinction, etc.) shall be awarded for Honors degree programme.

xi. Registration for Honors Degree program continue even if the student fails in any of the regular course after his registration to Honors Degree Course.

xii. A student who fails in any course of the Honor Degree Programme shall not be permitted to write the supplementary examination and the registration for the Honors Degree Programme will be cancelled.

xiii. If a student drops or is terminated from the Honors Degree program, the additional credits so far earned cannot be converted into open or core electives; they will remain extra. However, such students will receive a separate grade sheet mentioning the additional courses completed by them.

xiv. The Honors Degree will be mentioned in the degree certificate as Bachelor of Technology (Honors) in XYZ. For example, B.Tech. (Honors) in Mechanical Engineering.

xv. All the courses offered under Honors Degree may also be pursued through MOOCs. The Bloom's Taxonomy level for Honors Degree courses shall be 3 and above

Enrolment into Honors Degree:

i) Students of a Department/Discipline are eligible to opt for Honors Degree program offered by the same Department/Discipline

ii) The enrolment of student into Honors Degree is based on the CGPA obtained in the major degree program. CGPA shall be taken up to II B. Tech I semester in case of regular entry students and only II B. Tech I semester in case of lateral entry students. Students having 7 CGPA without any backlog subjects will be permitted to register for Honors Degree.

iii) If a student is detained due to lack of attendance either in Major Degree or in Honors Degree, registration for Honors Degree shall be cancelled.

iv) Transfer of credits from Honors Degree to regular B. Tech degree and vice-versa shall not be permitted.

v) Honors Degree is to be completed simultaneously with a Major degree program.

Registration for Honors / Minors:

i) The eligible and interested students shall apply through the HOD of his/her parent department. The whole process should be completed within one week before the start of every semester. Selected students shall be permitted to register the courses under Honors/Minors Degree.

ii) The students enrolled in the Honors/Minors Degree courses will be monitored continuously. An advisor/mentor from parent department shall be assigned to a group of students to monitor the progress.

iii) There shall be prescribed fee for registration of subjects for Honors/Minors Degree program offered in offline at the University.

Failure & Supplementary Examination Procedure for Honors/Minor Degree Courses:

i) Time Limit for Completion:
The student must successfully complete all Honors/ Minor Degree courses within the regular period of study applicable to the B. Tech. programme.

ii) Cancellation of Honors/ Minor Degree Registration:
If the student fails to complete all Honors / Minor Degree requirements within the stipulated regular period of study, the Honors/ Minor Degree registration shall stand cancelled, and the additional credits earned shall remain extra. These credits should not be counted for the award of the major Degree.

20. Attendance Requirements:

i) A student shall be eligible to appear for the University external examinations if he/she acquires a minimum of 40% attendance in each subject and 75% of attendance in aggregate of all the subjects. b) Condonation of shortage of attendance in aggregate up to 10% (65% and above and below 75%) in each semester may be granted by the Academic Council of the University.

ii) Shortage of Attendance below 65% in aggregate shall in NO CASE be condoned.

iii) A stipulated fee shall be payable towards condonation of shortage of attendance to the University.

iv) Students whose shortage of attendance is not condoned in any semester are not eligible to take their end examination of that class and their registration shall stand cancelled.

v) A student will not be promoted to the next semester unless he satisfies the attendance requirements of the present semester. They may seek readmission for that semester from the date of commencement of class work.

vi) If any candidate fulfils the attendance requirement in the present semester, he shall not be eligible for readmission into the same class.

vii) If the learning is carried out in blended mode (both offline & online), then the total attendance of the student shall be calculated considering the offline and online attendance of the student.

viii) For induction programme attendance shall be maintained as per AICTE norms.

ix) Attendance in SWAYAM courses shall be excluded from the calculation of the student's overall attendance requirement for the semester.

21. Conduct of Semester End Examination and Evaluation:

i) Semester end examination shall be conducted by the Controller of Examination (COE) by inviting Question Papers from the External / Internal Subject Experts. Director of Evaluation will decide the External / Internal subject experts.

ii) The answer papers of semester end examination should be evaluated externally / internally.

iii) The marks for the internal evaluation components will be added to the external evaluation marks secured in the Semester – End examinations, to arrive at total marks for any subject in that semester.

iv) Performance in all the subjects is tabulated program-wise and will be scrutinized by the office of the Controller of Examinations. Total marks obtained in each subject are converted into letter grades. Finally subject-wise marks and grades details,

subject-wise and branch-wise pass percentages are calculated.

22. Results Committee

i) Results Committee comprising of Vice Chancellor/nominee of Vice Chancellor, Director of Evaluation, Controller of Examinations, Dean of the School and One Senior Professor nominated by the Vice Chancellor on rotation basis will oversee the details of marks, grades and pass percentages of all the subjects and branch-wise pass percentages.

ii) Office of the Controller of Examinations will generate student-wise result sheets and the same will be published through University website.

iii) Student-wise Grade Sheets are generated and issued to the students.

23. Promotion Rules:

The following academic requirements must be satisfied in addition to the attendance requirements mentioned in section 20.

i) A student shall be promoted from first year to second year if he/she fulfils the minimum attendance requirement as per university norms.

ii) A student will be promoted from II B. Tech to III B. Tech if he/she fulfils the academic requirement of securing 40% of the credits (any ***decimal*** fraction should be ***rounded off*** to ***lower*** digit) up to in the subjects that have been studied up to II B. Tech I Semester.

iii) A student shall be promoted from III B. Tech to IV B. Tech if he/she fulfils the academic requirements of securing 40% of the credits (any ***decimal*** fraction should be ***rounded off*** to ***lower*** digit) in the subjects that have been studied up to III B. Tech I Semester.

And in case a student is detained for want of credits for a particular academic year by ii) & iii) above, the student may make up the credits through supplementary examinations and only after securing the required credits he/she shall be permitted to join in the III B. Tech I Semester or IV B. Tech I Semester respectively as the case may be.

iv) When a student is detained due to lack of credits/shortage of attendance he/she may be re-admitted when the semester is offered after fulfilment of academic regulations. In such case, he/she shall be in the Academic Regulations into which he/she is readmitted.

24. Grading:

As a measure of the student's performance, a 10-point Absolute Grading System using the following Letter Grades and corresponding percentage of marks shall be followed:
After each course is evaluated for 100 marks, the marks obtained in each course will be converted to a corresponding letter grade as given below, depending on the range in which the marks obtained by the student fall.

Structure of Grading of Academic Performance

Range in which the marks in the subject fall	Grade	Grade points Assigned
90 &above	S (Superior)	10
80-89	A (Excellent)	9
70-79	B (Very Good)	8
60-69	C (Good)	7
50-59	D (Average)	6
40-49	E (Below Average)	5
< 40	F (Fail)	0
Absent	Ab (Absent)	0

i) A student obtaining Grade 'F' or Grade 'Ab' in a subject shall be considered failed and will be required to reappear for that subject when it is offered the next supplementary examination.

ii) For non-credit audit courses, "Satisfactory" or "Unsatisfactory" shall be indicated instead of the letter grade and this will not be counted for the computation of

SGPA/CGPA/Percentage.

Computation of Semester Grade Point Average (SGPA) and Cumulative Grade Point Average (CGPA):

The Semester Grade Point Average (SGPA) is the ratio of sum of the product of the number of credits with the grade point scored by a student in all the courses taken by a student and the sum of the number of credits of all the courses undergone by a student, i.e.,

$$SGPA=\Sigma(C_i \times G_i)/\Sigma C_i$$

Where, "C_i" is the number of credits of the i^{th} subject and "G_i" is the grade point scored by the student in the i^{th} course.

The Cumulative Grade Point Average (CGPA) will be computed in the same manner considering all the courses undergone by a student overall the semesters of a program, i.e.,

$$CGPA=\Sigma(C_i \times S_i)/\Sigma C_i$$

Where "S_i" is the SGPA of the i^{th} semester and "C_i" is the total number of credits up to that semester.

Both SGPA and CGPA shall be rounded off to 2 decimal points and reported in the Grade Sheets / transcripts.

While computing the SGPA the subjects in which the student is awarded Zero grade points will also be included.

Grade Point: It is a numerical weightage allotted to each letter grade on a 10-point scale.

Letter Grade: It is an index of the performance of students in a said course. Grades are denoted by the letters S, A, B, C, D, E and F.

25. Grade sheet:

A grade sheet (Marks Memorandum) will be issued to each student Indicating his performance in all subjects registered in that semester indicating the SGPA and CGPA. SGPA and CGPA will be rounded off to the second place of decimal.

26. Consolidated Grade Sheet:

After successful completion of the entire Program of study, a Consolidated Grade Sheet containing performance of all semesters will be issued as a final record. Duplicate Consolidated Grade Sheet will also be issued, if required, after payment of requisite fee.

27. Award of Class: After a student has satisfied the requirements prescribed for the completion of the program and is eligible for the award of B. Tech. Degree, he/she shall be placed in one of the following four classes:

Class Awarded	CGPA Secured
First Class with Distinction	≥7.5
First Class	≥6.5<7.5
Second Class	≥5.5 <6.5
Pass Class	≥4.5 <5.5

Both SGPA, CGPA and Marks shall be given in the Grade Sheets.

28. Personal Verification /Recounting / Revaluation / Final Valuation

28.1 Personal Verification of Answer Scripts:

Candidates appear in a particular semester end-examinations may appeal for verification of their answer script(s) for arithmetic correction in totaling of marks and any omission / deletion in evaluation within 7 days from the date of declaration of results at the office of the Controller of Examinations on the prescribed proforma and by paying the prescribed fee per answer script.

It is clarified that personal verification of answer script shall not tantamount to revaluation of answer script. This is only a process of reverification by the candidate. Any mistake / deficiency with regard to arithmetic correction in totaling of marks and any omission / deletion in evaluation if found, the institution will correct the same.

28.2 Recounting / Revaluation:

Students shall be permitted for request for recounting/revaluation of the Semester-End examination answer scripts within a stipulated period after payment of prescribed fee. After recounting or revaluation, records are updated with changes if any and the student will be issued a revised grade sheet. If there are no changes, the same will be intimated to the students.

28.3 Final Valuation:

Students shall be permitted for request for final valuation of the Semester–End Examination answer scripts within a stipulated period after the publication of the revaluation results by paying the necessary fee. The final valuation shall be carried out by an expert not less than Associate Professor as per the scheme of valuation supplied by the examination branch in the presence of the student, Controller of Examinations and Director of Evaluation. However, students are not permitted to discuss / argue with the examiner. If the increase in marks after final valuation is equal to or more than 15% of the previous valuation marks, the marks obtained after final valuation shall be treated as final. If the variation of marks after final valuation is less than 15% of the previous valuation marks, then the earlier valuation marks shall be treated as the final marks.

29. With–holding of Results:

If the candidate has any case of indiscipline pending against him, the result of the candidate shall be withheld, and he will not be allowed/promoted into the next higher semester. The issue of degree is liable to be withheld in such cases.

30. Multiple Entry / Exit Option

(a) Exit Policy:

The students can choose to exit the four-year programme at the end of first / second / third year.

i) **UG Certificate in (Field of study/discipline)** - Programme duration: First year (first two semesters) of the undergraduate (B. Tech) programme, followed by an additional exit 10-credit bridge course(s) lasting two months, including at least 6-credit job-specific internship / apprenticeship that would help the candidates acquire job-ready competencies required to enter the work force.

ii) **UG Diploma (in Field of study/discipline)** - Programme duration: First two years (first four semesters) of the undergraduate (B. Tech) programme, followed by an additional exit 10-credit bridge course(s) lasting two months, including at least 6-creditjob-specific internship / apprenticeship that would help the candidates acquire job-ready competencies required to enter the workforce.

iii) **Bachelor of Science (in Field of study / discipline)** i.e., **B. Sc. Engineering in (Field of study / discipline)**- Programme duration: First three years (first six semesters) of the undergraduate (B. Tech) programme.

(b) Entry Policy:

Modalities on multiple entry by the student into the B. Tech. programme will be provided in due course of time.

Note:

The University shall resolve any issues that may arise in the implementation of Multiple Entry and Exit policies from time to time and shall review the policies in the light of periodic changes brought by UGC, AICTE and State Government.

31. Gap Year Concept:

Gap year concept for Student Entrepreneur in Residence is introduced and outstanding students who wish to pursue entrepreneurship/ become entrepreneur are allowed to take a break of one year at any time after II B. Tech to pursue full-time entrepreneurship programme / to establish start-ups. This period may be extended to two years at the most and these two years would not be counted for the time for the maximum time for graduation. The Head of the respective Department shall forward such proposals submitted by the students to the University. An evaluation committee constituted by the University shall evaluate the proposal submitted by the student and the committee shall decide whether to permit the student(s) to avail the Gap Year or not.

32. Transitory Regulations

Discontinued, detained, or failed candidates are eligible for readmission as and when the semester is offered after fulfilment of Academic Regulations. Candidates who have been detained for want of attendance or not fulfilled academic requirements or who have failed after having undergone the course in earlier regulations or have discontinued and wish to continue the course are eligible for admission into the unfinished semester from the date of commencement of classwork with the same or equivalent subjects as and when subjects are offered, and they will follow the academic regulations into which they are readmitted.

Candidates who are permitted to avail Gap Year shall be eligible for rejoining into the succeeding year of their B. Tech from the date of commencement of class work, and they will follow the academic regulations into which they are readmitted

33. Minimum Instructional Days for a Semester:

The minimum instructional weeks excluding exams for each semester shall be 15 weeks.

34. Medium of Instruction

The medium of instruction of the entire B. Tech undergraduate programme in Engineering & Technology (including examinations and project reports) will be in English only.

35. Student Transfers:

Student transfers shall be as per the guidelines issued by the Government of Andhra Pradesh and the University from time to time.

36. Re-registration for Improvement of Internal Evaluation Marks:

A candidate shall be given one chance to re-register for each subject provided the internal marks secured by a candidate are less than 50% and has failed in the end examination.

36.1 The candidate should have completed their four years of course work and obtained examinations results for all the four years and are permitted to re-register as and when the respective subjects are offered.

36.2 Out of the subjects the candidate has failed in the examination due to Internal Evaluation marks secured being less than 50%, the candidate shall be given one chance for each Theory subject and for a maximum of ten Theory subjects for Improvement of Internal evaluation marks.

36.3 The candidate has to re-register for the chosen subjects and fulfill the academic requirements.

36.4 For re-registration the candidates have to submit the applications to the Dean of the Concerned School through the Head of the Department by paying the requisite fees (For each subject, the candidate has to pay a requisite fee and the amount is to be remitted in the form of D.D./ Challan in favour of the Registrar, RVS University) and get approval from the Dean of the Concerned School before the start of the semester in which re-registration is required.

36.5 In the event of availing the Improvement of Internal evaluation marks, the internal evaluation marks as well as the End Examinations marks secured in the previous attempt(s) for the re-registered subjects stand cancelled. The internal marks obtained by the students in the re-registration shall be treated as final.

37. General Instructions:

37.1 The academic regulations should be read as a whole for purpose of any interpretation.

37.2 Malpractices rules nature and punishments are appended

37.3 Where the words "he", "him", "his", occur in the regulations, they also include "she", "her", "hers", respectively.

37.4 The University may change or amend the academic regulations or syllabi at any time and the changes or amendments shall be made applicable to all the students on rolls with effect from the dates notified by the University.

37.5 In the case of any doubt or ambiguity in the interpretation of the above rules, the decision of the Vice-Chancellor is final.

FAILURE TO READ AND UNDERSTAND THE REGULATIONS IS NOT AN EXCUSE

RVSU Academic Regulations-R26 for B. Tech (Lateral Entry Scheme)

(Effective for the students admitted into II year B. Tech through Lateral Entry Scheme from the Academic Year **2027-28** onwards)

1. Award of the Degree

(a) Award of the B. Tech. Degree / B. Tech. Degree with a Minor if he/she fulfils the following:

i. Pursues a course of study for not less than three academic years and not more than six academic years. However, for the students availing Gap year facility this period shall be extended by two years at the most and these two years would in addition to the maximum period permitted for graduation (Six years).

ii) Registers for 127 credits and secures all 127 credits

(b) Award of B. Tech. degree with Honors if he / she fulfils the following:

i) Student secures additional 18 credits fulfilling all the requisites of a B. Tech. program i.e., 127 credits.

ii) Registering for Honors is optional.

iii) Honors is to be completed simultaneously with B.Tech. programme.

(c) Award of B.Tech. degree with Minor

i) Student secures additional 18 credits fulfilling all the requisites of a B.Tech. program i.e., 127 credits.

ii) Registering for Minor Degree is optional.

iii) Minor Degree is to be completed simultaneously with B. Tech. programme.

2. Students, who fail to fulfil the requirement for the award of the degree within six consecutive academic years from the year of admission, shall forfeit their seat.

3. Course Pattern

i) The entire course of study is three academic years on semester pattern.

ii) A student eligible to appear for the end examination in a subject but absent at it or has failed in the end examination may appear for that subject at the next supplementary examination offered.

iii) When a student is detained due to lack of credits / shortage of attendance the student may be re-admitted when the semester is offered after fulfilment of academic regulations, the student shall be in the academic regulations into which he/she is readmitted.

4. All other academic regulations, including those relating to attendance, examinations, promotion, evaluation, and the award of the degree, shall be the same as those prescribed for regular B. Tech. students, unless otherwise specified

Identification of Courses

B. Tech

Each course shall be uniquely identified by an alphanumeric code of width 7 characters as given below.

No. of Digits	Description
First two digits	Year of regulations Ex:26
Next one letter	**Type of program:** A: B. Tech B: M. Tech C: M.B.A D: M.C.A E: BBA F: BCA G: B. Sc (Hon) H: B. Sc (AHS)
Next two letters	**Code of program**: CE: Civil Engineering EE: Electrical and Electronics engineering ME: Mechanical Engineering EC: Electronics and Communications Engineering CS: Computer Science and Engineering CD: Computer Science and Engineering (Data Science) CM: Computer Science and Engineering (Artificial Intelligence and Machine Learning) IT: Information Technology CA: Computer Science and Engineering (Artificial Intelligence) CC: Computer Science and Engineering (Cyber Security) CO: Computer Science and Engineering (IOT) CB: Computer Science and Business Systems AI: Artificial Intelligence AD: Artificial Intelligence and Data Science CQ: Computer Science and Engineering (Quantum Computing) DE: Computer Science and Engineering (Artificial Intelligence and Data Engineering) AR: Computer Science and Engineering (Artificial Intelligence and Robotics) AC: Computer Science and Engineering (Cloud Computing and Automation) DT: Electronics and Communication Engineering with Specialization in Drone Technology
Last two digits	Indicate serial numbers: ≥ 01

Ex:

26ACE01; 26AEE01; 26AME01; 26AEC01; 26ACS01; 26ACD01; 26ACM01; 26AIT01; 26ACA01; 26ACC01; 26ACO01; 26ACB01; 26AAI01; 26AAD01; 26ACQ01; 26ADE01; 26AAR01; 26AAC01; 26ADT01; 26AHS01; 26AMB01

RULES FOR DISCIPLINARY ACTION FOR MALPRACTICES / IMPROPER CONDUCT IN EXAMINATIONS

	Nature of Malpractices/Improper conduct	Punishment
	If the candidate:	
1.(a)	Possesses or keeps accessible in examination hall, any paper, notebook, programmable calculators, Cell phones, pager, palm computers or any other form of material concerned with or related to the subject of the examination (theory or practical) in which he is appearing but has not made use of (material shall include any marks on the body of the candidate which can be used as an aid in the subject of the examination)	Expulsion from the examination hall and cancellation of the performance in that subject only.
(b)	Gives assistance or guidance or receives it from any other candidate orally or by any other body language methods or communicates through cell phones with any candidate or persons in or outside the exam hall in respect of any matter.	Expulsion from the examination hall and cancellation of the performance in that subject only of all the candidates involved. In case of an outsider, he will be handed over to the police and a case is registered against him.
2.	Has copied in the examination hall from any paper, book, programmable calculators, palm computers or any other form of material relevant to the subject of the examination (theory or practical) in which the candidate is appearing.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted to appear for the remaining examinations of the subjects of that semester/year. The Hall Ticket of the candidate is to be cancelled and sent to the University.
3.	Impersonates any other candidate in connection with the examination.	The candidate who has impersonated shall be expelled from examination hall. The candidate is also debarred for four consecutive semesters from class work and all University examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat. The performance of the original candidate who has been impersonated, shall be cancelled in all the subjects of the examination (including practicals and project work) already appeared and shall not be allowed to appear for examinations of the remaining subjects of that semester/year. The candidate is also debarred for four consecutive semesters from class work and all University examinations, if his involvement is established. Otherwise, the candidate is debarred for two consecutive semesters from class work and all University examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat. If the imposter is an outsider, he will be handed over to the police and a case is registered against him.

4.	Smuggles in the Answer book or additional sheet or takes out or arranges to send out the question paper during the examination or answer book or additional sheet, during or after the examination.	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The candidate is also debarred for two consecutive semesters from class work and all University examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat.
5.	Uses objectionable, abusive or offensive language in the answer paper or in letters to the examiners or writes to the examiner requesting him to award pass marks.	Cancellation of the performance in that subject only.
6.	Refuses to obey the orders of the Chief Superintendent /Assistant - Superintendent /any officer on duty or misbehaves or creates disturbance of any kind in and around the examination hall or organizes a walk out or instigates others to walk out, or threatens the officer-in charge or any person on duty in or outside the examination hall of any injury to his person or to any of his relations whether by words, either spoken or written or by signs or by visible representation, assaults the officer-in-charge, or any person on duty in or outside the examination hall or any of his relations, or indulges in any other act of misconduct or mischief which result in damage to or destruction of property in the examination hall or any part of the College campus or engages in any other act which in the opinion of the officer on duty amounts to use of unfair means or misconduct or has the tendency to disrupt the orderly conduct of the examination.	In case of students of the college, they shall be expelled from examination halls and cancellation of their performance in that subject and all other subjects the candidate(s) has (have) already appeared and shall not be permitted to appear for the remaining examinations of the subjects of that semester/year. If the candidate physically assaults the invigilator/ officer-in-charge of the Examinations, then the candidate is also debarred and forfeits his/her seat. In case of outsiders, they will be handed over to the police and a police case is registered against them.
7.	Leaves the exam hall taking away answer script or intentionally tears of the script or any part thereof inside or outside the examination hall.	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The candidate is also debarred for two consecutive semesters from class work and all University examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat.
8.	Possess any lethal weapon or firearm in the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The candidate is also debarred and forfeits the seat.

9.	If student of the college, who is not a candidate for the particular examination or any person not connected with the college indulges in any malpractice or improper conduct mentioned in clause 6 to 8.	Student of the colleges expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The candidate is also debarred and forfeits the seat. Person (s) who do not belong to the College will be handed over to police and, a police case will be registered against them.
10.	Comes in a drunken condition to the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year.
11.	Copying detected on the basis of internal evidence, such as, during valuation or during special scrutiny.	Cancellation of the performance in that subject only or in that subject and all other subjects the candidate has appeared including practical examinations and project work of that semester / year examinations, depending on the recommendation of the committee.
12	Tampering of the NPTEL / SWAYAM / SWAYAM certificate	Cancellation of student performance in the entire semester
13.	If any malpractice is detected which is not covered in the above clauses 1 to 12 shall be reported to the University for further action to award suitable punishment.	-

Note:-

Whenever the performance of a student is cancelled in any subject/subjects due to Malpractice, he has to register for End Examinations in that subject/subjects consequently and has to fulfil all the norms required for the award of Degree.

DEPARTMENT OF MECHANICAL ENGINEERING

VISION AND MISSION OF THE DEPARTMENT

VISION:

To become a centre of excellence in advanced mechanical engineering, nurturing innovators who transform design, manufacturing, automation and engineering systems through sustainable technologies.

MISSION:

M1: Building proficiency in engineering design, manufacturing, thermal systems, materials and industrial processes among learners through experiential learning.

M2: Nurturing research, innovation and entrepreneurship in advanced manufacturing, automation, robotics and additive technologies.

M3: Bridging academic knowledge with manufacturing practice through industrial training, live projects and professional collaboration.

M4: Advancing resource-efficient products and processes responsive to industrial, environmental and societal needs.

M5: Enhancing faculty expertise and providing advanced workshops, laboratories, design tools and manufacturing facilities.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

PEO1: Graduates will pursue higher education and advanced research in mechanical engineering, manufacturing, thermal sciences, design, automation and interdisciplinary engineering domains.

PEO2: Graduates will establish successful careers by applying engineering principles to the design, analysis, manufacturing, operation and maintenance of mechanical systems and industrial processes.

PEO3: Graduates will identify technological and market opportunities and develop innovative products, manufacturing solutions, engineering services and technology-based enterprises.

PEO4: Graduates will continuously acquire knowledge of advanced manufacturing, automation, robotics, digital engineering, sustainable technologies and emerging industrial practices.

PROGRAMME SPECIFIC OUTCOMES (PSOs)

On successful completion of the Program, the graduates of B. Tech. Mechanical Engineering will be able to:

PSO1: Apply principles of engineering mechanics, thermodynamics, fluid mechanics, heat transfer, materials science and machine theory to analyse, model and solve complex mechanical and thermal engineering problems.

PSO2: Design, develop and manufacture mechanical components, machines, thermal systems and production systems using CAD/CAM, computational and simulation tools, automation, advanced manufacturing technologies and sustainable engineering practices.

PROGRAM OUTCOMES (POs)

PO1 - Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization.

PO2 - Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

PO3 - Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

PO4 - Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.

PO5 - Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

PO6 - The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

PO7 - Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.

PO8 - Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9 - Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO10 - Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11 - Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

RVS UNIVERSITY

FIND YOUR PLACE | FOLLOW YOUR PASSION

Estd. under Andhra Pradesh Private Universities (Establishment and Regulation) Act, 2016

NAAC A+ GRADE ACCREDITED UNIVERSITY

B.TECH. - COURSE STRUCTURE – R26
(Applicable from the academic year 2026-27 onwards)

INDUCTION PROGRAMME
(0 Semester – 3 weeks duration)

Sl. No.	Activities
1.	Physical Activities - Sports, Yoga and Meditation, Plantation
2.	Career Counselling
3.	Orientation to all branches - Career Options, Tools,etc.
4.	Orientation on admitted branch - Corresponding Labs, Tools and Platforms
5.	Proficiency Modules & Productivity Tools
6.	Assessment on Basic Aptitude and Mathematical Skills
7.	Remedial Training in Foundation Courses
8.	Human Values & Professional Ethics
9.	Communication Skills - focus on Listening, Speaking, Reading, Writing skills
10.	Concepts of Programming
11.	Mentor-Mentee Groups Meet
12.	Visits in Local Area
13.	Lectures & Workshops by Eminent People
14.	Feedback and Report on the Program

DEPARTMENT OF MECHANICAL ENGINEERING

Course Structure & Scheme of Examination

I B. Tech I Semester – ME **Academic Regulations: R26**

S. No	Category	Course Code	Course Name	Hours/Week			Credits	Scheme of Examination (Maximum Marks)		
				L	T	P	C	CIA	SEE	Total
1.	HS	26AHS01	Communicative English	2	0	2	3	80	120	200
2.	BS	26AHS02	Engineering Chemistry	3	0	2	4	80	120	200
3.	BS	26AHS03	Linear Algebra and Calculus	3	0	0	3	40	60	100
4.	PC	26ACS01	Computational Thinking and Problem Solving with Python	3	0	2	4	80	120	200
5.	ES	26AAI01	AI Tools and Applications	1	0	4	3	40	60	100
6.	ES	26AME01	Engineering Workshop	0	0	3	1.5	40	60	100
7.	HS	26AHS08	Health and Wellness, Yoga and Sports	0	0	1	0.5	100	-	100
TOTAL				**12**	**0**	**14**	**19**	**460**	**540**	**1000**

I B. Tech II Semester – ME **Academic Regulations: R26**

S. No	Category	Course Code	Course Name	Hours/Week			Credits	Scheme of Examination (Maximum Marks)		
				L	T	P	C	CIA	SEE	Total
1.	BS	26AHS04	Engineering Physics	3	0	2	4	80	120	200
2.	ES	26ACE01	Engineering Mechanics	3	0	2	4	80	120	200
3.	BS	26AHS07	Differential Equations and Vector Calculus	3	0	0	3	40	60	100
4.	ES	26ACS02	Data Structures Using Python	3	0	2	4	80	120	200
5.	ES	26AEE02	Basic Electrical and Electronics Engineering	3	0	0	3	40	60	100
6.	ES	26AME02	Engineering Graphics	1	0	4	3	40	60	100
7.	ES	26AME03	Basic Simulation Lab	0	0	3	1.5	40	60	100
8.	HS	26AHS09	NSS/ NCC/ Scouts and Guides	0	0	1	0.5	100	-	100
TOTAL				**16**	**0**	**14**	**23**	**500**	**600**	**1100**

I B. Tech I Semester - Common to CIVIL, EEE, ME, CSE, CSD, CSM, CSE (AI&DE), CSBS and CSE (AI & R)
I B. Tech II Semester - Common to ECE, IT, CAI, CSC, CSO, AI, AI & DS, ECE (DT), CSE (CCA) and CSE (QC)

L	T	P	C
2	0	2	3

26AHS01: COMMUNICATIVE ENGLISH

Course Outcomes (COs):

After successful completion of the course, students will be able to:

CO1 : Identify the context, main idea and specific information from social, academic and transactional dialogues and short audio texts.

CO2 : Apply appropriate grammatical structures, vocabulary related to Engineering and contextual language conventions to construct and edit grammatically correct sentences and comprehend academic and technical texts.

CO3 : Participate effectively in pair and group discussions by organizing ideas, using appropriate discourse markers, and communicating clearly in formal and informal academic contexts.

CO4 : Write clear, grammatically correct and contextually appropriate academic and professional documents, including official letters, technical reports and resumes.

CO5 : Demonstrate fluency, clarity and confidence in academic and professional oral communication through structured presentations, extempore speaking and conversational activities.

UNIT- I: HUMAN VALUES: GIFT OF MAGI (SHORT STORY) 6 Hours

Listening : Identifying the topic, the context and specific pieces of information by listening to short audio texts and answering a series of questions.

Speaking : Asking and answering general questions on familiar topics such as home, family, work, studies and interests and introducing oneself and others.

Reading : Skimming to get the main idea of a text; scanning to look for specific piece of information.

Writing : Mechanics of Writing-Capitalization, Spellings, Punctuation-Parts of Sentences.

Grammar : Parts of Speech, Basic Sentence Structures-forming questions

Vocabulary	:	Synonyms, Antonyms, Affixes (Prefixes/Suffixes), Root words.
AI Intervention	:	Grammarly, Merriam-Webster Vocabulary Building, Gamma.app

UNIT – II: NATURE: THE BROOK BY ALFRED TENNYSON (POEM)

5 Hours

Listening	:	Answering a series of questions about main ideas and supporting ideas after listening to audio text.
Speaking	:	Discussion in pairs / small groups on specific topics followed by short structural talks.
Reading	:	Identifying sequence of ideas; recognizing verbal techniques that help to link the ideas in a paragraph together
Writing	:	Structure of a paragraph - Paragraph writing (specific topics)
Grammar	:	Cohesive devices -linkers, use of articles and zero article; prepositions
Vocabulary	:	Homonyms, Homophones, Homographs.
AI Intervention	:	ELSA Speak, ChatGPT / Google Gemini, Merriam-Webster Vocabulary Building

UNIT – III: BIOGRAPHY: ELON MUSK 7 Hours

Listening	:	Listening for global comprehension and summarizing what is listened to.
Speaking	:	Discussing specific topics in pairs or small groups and reporting what are discussed.
Reading	:	Reading a text in detail by making basic inferences -recognizing and interpretingspecific context clues; strategies to use text clues for comprehension.
Writing	:	Summarizing, Note-making, paraphrasing
Grammar	:	Verbs - tenses; subject-verb agreement; Compound words, Collocations
Vocabulary	:	Compound words, Collocations
AI Intervention	:	Grammarly, ELSA Speak, Speech-to-Text, SmallTalk2Me

UNIT- IV: INSPIRATION: THE TOYS OF PEACE BY SAKI 7 Hours

Listening	:	Making predictions while listening to conversations/ transactional dialogues without video; listening with video
Speaking	:	Role plays for practice of conversational English in academic contexts (formal and informal) - asking for and giving information / directions
Reading	:	Studying the use of graphic elements in texts to convey

		information, reveal trends / patterns / relationships, communicate processes or display complicated data.
Writing	:	Letter Writing: Official Letters, Technical Report, Resumes
Grammar	:	Reporting verbs, Direct & Indirect speech, Active & Passive Voice
Vocabulary	:	Words often confused, Jargons
AI Intervention	:	Real-time grammar, ELSA Speak, Speech-to-Text

UNIT- V: MOTIVATION: THE POWER OF INTRAPERSONAL COMMUNICATION (AN ESSAY) 5 Hours

Listening	:	Identifying key terms, understanding the concepts and answering a series of relevant questions that test comprehension
Speaking	:	Formal oral presentations on topics from academic contexts
Reading	:	Reading comprehension
Writing	:	Critical Writing - Writing structured essays on specific topics.
Grammar	:	Editing short texts –identifying and correcting common errors in grammar and usage (articles, prepositions, tenses, subject verb agreement)
Vocabulary	:	Technical Jargons
AI Intervention	:	Notion AI, ELSA Speak, Speech-to-Text, Gamma.app

LIST OF EXERCISES:

Exercise - 1: Vowels & Consonants
Exercise - 2: Neutralization/Accent Rules
Exercise - 3: Communication Skills & JAM
Exercise - 4: Role Play or Conversational Practice
Exercise - 5: E-mail Writing & Technical Report Writing
Exercise - 6: Cover Letter, Resume Writing & SOP
Exercise - 7: Group Discussions-methods & practice
Exercise - 8: Debates - Methods & Practice
Exercise - 9: PPT Presentations/ Poster Presentation
Exercise - 10: Interviews Skills

TEXT BOOKS:

1. *Pathfinder: Communicative English for Undergraduate Students*, 1st Edition, Orient Black Swan, 2023.
2. *Empowering with Language*, 1st Edition, Cengage Learning India Pvt. Ltd., 2023.

REFERENCE BOOKS:

1. Dubey, Sham Ji & Co. *English for Engineers*. Vikas Publishers, 2021.
2. Lewis, Norman. *Word Power Made Easy- The Complete Handbook for*

Building aSuperior Vocabulary. Penguin and Goyal, 2026.
3. Wren P.C. and Martin H., *High School English Grammar and Composition*, S. Chand, Revised Edition, 2026.
4. J. Sethi & P.V. Dhamija. *A Course in Phonetics and Spoken English*, 2nd Edition, Phi learning, 2023.

ONLINE RESOURCES:

1. British Council – Learn English – https://learnenglish.britishcouncil.org
2. NPTEL – Communication Skills – https://nptel.ac.in/courses/109104101
3. Purdue OWL (Online Writing Lab) – https://owl.purdue.edu
4. BBC Learning English – https://www.bbc.co.uk/learningenglish
5. TED Talks – https://www.ted.com/talks (Listening & speaking practice)
6. Coursera English Communication Skills Specialization – https://www.coursera.org
7. EdX English for Professional and Academic Purposes – https://www.edx.org
8. ELSA Speak (App) – https://elsaspeak.com (AI pronunciation coach)

LAB SOFTWARE:

- Young India Films
- Walden InfoTech

Course Articulation Matrix:

Course Outcomes	Program Outcomes										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	2	–	–	1	–	–	2	3	–	–
CO2	3	2	1	1	2	–	–	2	3	–	–
CO3	2	3	1	1	1	–	–	3	3	–	–
CO4	2	3	2	2	2	–	–	2	3	–	–
CO5	2	2	1	1	2	–	–	3	3	–	–

Correlation Levels: 3- High; 2-Moderate; and 1- Low

I B.Tech I Sem - Common to CE and ME

L	T	P	C
3	0	2	4

26AHS02: ENGINEERING CHEMISTRY

Course Outcomes:

After successful completion of the course, students will be able to:

CO1 : Analyze corrosion mechanisms in engineering materials and recommend suitable corrosion prevention and protection techniques.

CO2 : Evaluate the properties of lubricants, fuels and water and select suitable materials or treatment methods for engineering applications.

CO3 : Select and justify suitable polymers, ceramics, refractories and composite materials for specific engineering applications.

CO4 : Analyze the properties and applications of cement, alloys, conductors, semiconductors and insulators for appropriate engineering applications.

CO5 : Apply green chemistry and sustainable engineering principles to evaluate materials, processes and environmental impacts.

UNIT- I: ELECTROCHEMISTRY AND CORROSION 9 Hours

Electrochemical cells (Galvanic and electrolytic cells); primary batteries: construction, working and electrical applications of zinc-air battery; secondary batteries: construction, reactions and applications of lead-acid, lithium-ion batteries; supercapacitors: definition, classification and applications; electric fuel cell: construction, working and electrical applications of hydrogen-oxygen fuel cells, Chemical and Electrochemical theories of corrosion: anodic and cathodic reactions, electrochemical series. Types of corrosion: galvanic, pitting, crevice, inter granular, stress corrosion. Electroplating: principles - Copper and Nickel plating.

AI Integration: Machine learning models for corrosion prediction based on environmental and material data. Image - based corrosion detection using computer vision. AI models for battery health monitoring and energy optimization in IoT devices. Predictive analytics for battery's life and performance.

UNIT - II: CHEMISTRY OF LUBRICANTS, FUELS AND WATER 9 Hours

Functions of lubricants, classification and applications of Solid lubricants, Semi-solid lubricants, Liquid lubricants. Properties of lubricants (Definition and significance). Classification of fuels: solid, liquid, gaseous fuels, calorific value: HCV and LCV, Calorific value determination using bomb calorimeter, problems. Green fuels - Green hydrogen, Hythane. Combustion - problems. Water chemistry: hardness of water, temporary and permanent hardness of water, hardness units, problems, Estimation of hardness of water by EDTA method, Boiler troubles: scale and sludge formation, boiler

corrosion, caustic embrittlement.

AI Integration: Predictive maintenance using AI for lubricant degradation analysis. Data-driven selection of lubricants based on operating conditions. AI models to optimize fuel efficiency and combustion performance. Data analysis of engine parameters using regression models. Prediction of water quality trends using Python (Pandas, Matplotlib). Regression models for forecasting BOD/COD. Visualization of water treatment efficiency using datasets.

UNIT- III: POLYMERS AND COMPOSITES 9 Hours

Polymer chemistry: types of polymerizations, examples, thermoplastics, thermosets, Engineering polymers: Preparation and engineering applications of Nylon, Acrylonitrile Butadiene styrene (ABS), Polytetrafluoroethylene (PTFE). Elastomers- Natural rubber, synthetic rubbers (BUNA-S and BUNA-N). Ceramics: alumina, silicon carbide, silicon nitride, zirconia and applications Refractories: characteristics, classification and properties of refractories. Composite materials: classification and applications.

AI Integration: Computer vision for polymer structure identification and degradation analysis. Image-based classification of materials. AI-based material property prediction and selection. Machine learning for composite optimization and performance analysis.

UNIT - IV: ENGINEERING MATERIALS 9 Hours

Special Types of cement and their applications. A. High alumina cements B. High early strength cements (HES) C. Silicate cements D. Sorel cements. Steel chemistry: composition and applications of steel, alloying elements, Heat treatment: annealing, normalizing, hardening, tempering, age hardening. Chemical admixtures: plasticizers, retarders, accelerators, air-entraining agents. Introduction, classification and applications of conductors - copper, aluminium, silver, alloys (brass, bronze), semiconductors (Si, Ge, GaAs), doping -- n-type (P, As) and p-type (B, Al); superconductors, insulators: definition, classification - solid insulators (ceramics, porcelain,), liquid insulators (transformer oil and synthetic esters), gaseous insulators (air, SF6, nitrogen)

AI Integration: Prediction of concrete compressive strength using machine learning models. Data analysis of mix design parameters. Material selection using ML (property prediction, multi-objective optimization); defect classification in insulation using image analytics.

UNIT- V: CHEMISTRY OF GREEN ENGINEERING 9 Hours

Principles of green chemistry and their applications. Sustainable construction materials: geopolymer concrete, recycled aggregates, industrial by-products (fly ash, Ground Granulated Blast-furnace Slag (GGBS) Air pollution: types, sources, effects; air quality index, Water pollution: sources, effects and remedial methods. Chemistry of greenhouse gases and climate changes, Soil chemistry: pH, organic matter, contamination, remediation. Solid waste management: classification, treatment, recycling in construction

AI Integration: Life Cycle Analysis (LCA) using AI tools. Air quality prediction models using machine learning. Sustainability assessment of materials.

LIST OF EXPERIMENTS:

1. Estimation of Ferrous Ion (Fe^{+2}) using potassium dichromate ($K_2Cr_2O_7$) by Redox titration.
2. Determination of rate of corrosion by weight loss method.
3. Estimation of Total hardness, permanent and temporary hardness of water by using Ethelene diamine tetra acetic acid (EDTA) by complexometric titration.
4. Determination of Total alkalinity (due to OH^- and CO_3^{2-}) of water using sodium hydroxide (NaOH) (acid-base titration)
5. Estimation of Dissolved Oxygen in water sample. (Iodometry)
6. Determination of strength of acid (HCl) using Sodium Hydroxide
7. Determination of moisture content in coal sample (Proximate analysis)
8. Conductometric titrations of strong acid Vs strong base (Ex: HCl Vs NaOH)
9. Determination of strength of Hydrochloric acid (HCl) using Sodium Hydroxide (NaOH) by pH metry
10. Preparation of Polymer (Phenol-Formaldehyde →Bakelite)
11. Estimation of amount of Iron (Fe^{+2}) using potassium dichromate ($K_2Cr_2O_7$) by Potentiometry
12. Determination of viscosity of lubricating oil using Red-Wood Viscometer – I

TEXT BOOKS:

1. P. C. Jain & M. Jain, *Engineering chemistry*, 17th Edition, Dhanpat Rai Publishing Company, 2025.
2. S. S. Dara & S.S. Umare, *Engineering chemistry*, 12th Edition, S. Chand Publishing, 2022.

REFERENCE BOOKS:

1. M. L. Davis, *Water and wastewater engineering*: Design principles and practice, 2nd Edition, McGraw-Hill Education, 2020.
2. Prasanta Rath & S. Aruna Kumari, Engineering Chemistry, 1st Edition, Cengage, 2024
3. R. N. Goyal & Harmendra Goel, *Textbook of Engineering Chemistry*, 3rd Edition, Springer 2026

ONLINE RESOURCES:

1. https://ec-iitr.vlabs.ac.in/exp/water-hardness/
2. https://ec-iitr.vlabs.ac.in/exp/water-alkalinity/
3. https://inoc-amrt.vlabs.ac.in/exp/acid-base-titration/simulation.html
4. Python (Scikit-learn, NumPy, Pandas, Matplotlib)
5. Orange Data Mining
6. Open LCA
7. KNIME / Teachable Machine

Course Articulation Matrix:

Course Outcomes	Program Outcomes										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2	2	–	–	–	1	–	–
CO2	3	3	3	2	2	–	–	–	1	–	–
CO3	3	3	3	2	2	–	–	1	2	–	–
CO4	3	3	3	2	2	–	–	–	1	–	–
CO5	3	3	3	3	2	–	–	1	2	–	–

Correlation Levels: 3-High; 2-Moderate; and 1-Low

I B.Tech. I Sem- Common to All Branches

L	T	P	C
3	0	0	3

26AHS03: LINEAR ALGEBRA AND CALCULUS

Course Outcomes

After Successful completion of the course, the student will be able to:

CO1 : Apply matrix algebra techniques to analyze matrix properties and solve systems of linear equations using direct and iterative methods in engineering applications.

CO2 : Analyze Eigen value and eigenvector problems, apply diagonalization and Cayley- Hamilton theorem, reduce and classify quadratic forms using appropriate transformations.

CO3 : Apply mean value theorems, Taylor and Maclaurin series, curvature concepts to analyze and approximate engineering functions and curves.

CO4 : Analyze multivariable functions using partial and total derivatives, Jacobians and directional derivatives, solve unconstrained and constrained optimization problems.

CO5 : Evaluate double and triple integrals using appropriate coordinate transformations to determine areas, volumes, mass, centroids and moments in engineering applications.

UNIT – I: MATRIX ALGEBRA AND LINEAR SYSTEMS 9 Hours

Matrices, types of matrices, operations, transpose, inverse, symmetric and skew-symmetric matrices; determinant properties; rank of a matrix; echelon and normal forms; system of linear equations; consistency conditions; Gauss elimination and Gauss-Jordan methods; iterative methods: Jacobi and Gauss-Seidel; applications to engineering systems.

AI Integration: Python-based solution of large linear systems using NumPy; visualization of convergence in iterative methods; AI-assisted debugging of matrix algorithms; using ChatGPT/Copilot to explain row operations and rank reduction.

UNIT - II: EIGEN VALUES, EIGEN VECTORS, DIAGONALIZATION & QUADRATIC FORMS 9 Hours

Eigen values and eigenvectors; characteristic equation; Cayley-Hamilton theorem; diagonalization; similarity transformation; powers and inverse of matrices using Cayley-Hamilton theorem; quadratic forms; canonical form; reduction by orthogonal transformation; rank, index, signature; positive definite forms.

AI Integration: AI-assisted Eigen value computation and verification; Python/SciPy visualization of Eigen modes and quadratic surfaces; use of AI tools to explain diagonalization and matrix classification; numerical checking of Cayley-Hamilton theorem.

UNIT - III: SINGLE VARIABLE CALCULUS AND SERIES EXPANSIONS

9 Hours

Rolle's mean value theorem; Lagrange's mean value theorem; Cauchy's mean value theorem; Taylor's and Maclaurin's series with remainder; approximation of functions; curvature of plane curves; radius and centre of curvature; circle of curvature; applications to engineering approximations and curve design.

AI Integration: SymPy-based Taylor and Maclaurin expansion generation; plotting truncation error and convergence; AI-supported derivation checking for mean value theorems; curvature plot visualization and interpretation.

UNIT - IV: MULTI VARIABLE DIFFERENTIATION AND OPTIMIZATION

9 Hours

Functions of several variables: Continuity and Differentiability, Partial derivatives, total derivatives, chain rule, Directional derivative, Taylor's and Maclaurin's series expansion of functions of two variables. Jacobian, Functional dependence, maxima and minima of functions of two variables, method of Lagrange multipliers.

AI Integration: Gradient field visualization using Python; AI-assisted Jacobian computation and verification; constrained optimization using SciPy; use of AI tools to interpret saddle points, extrema, and optimization constraints.

UNIT-V: MULTIPLE INTEGRALS AND COORDINATE TRANSFORMATIONS

9 Hours

Double integrals; change of order of integration; triple integrals; change of variables; polar, cylindrical and spherical coordinates; area and volume computation; mass, centroid and moment calculations; engineering applications of multiple integration.

AI Integration: Numerical evaluation of double and triple integrals using SciPy; 2D and 3D region visualization; AI-assisted generation of integration code; interpreting Jacobian as geometric scaling factor in transformed coordinates.

TEXT BOOKS:

1. E. Keryszig, *Advanced Engineering Mathematics*, 11th Edition, International Adaptation, Hoboken, NJ, USA: Wiley, 2025.
2. B. S. Grewal, *Higher Engineering Mathematics*, 45th Edition, New Delhi, India: Khanna Publishers, 2024.
3. J. R. Hass, C. E. Heil, M. D. Weir, and P. Bogacki, *Thomas' Calculus*, 15th Edition, Pearson, 2022.
4. R. K. Jain and S. R. K. Iyengar, *Advanced Engineering Mathematics*, 5th Edition, Oxford, U.K.: Alpha Science International, 2021.

REFERENCE BOOKS:

1. G. Strang, *Introduction to Linear Algebra*, 6th Edition, Wellesley Cambridge Press, 2023, ISBN: 978-1-7331466-7-8.
2. D. C. Lay, S. R. Lay, and J. J. McDonald, *Linear Algebra and Its Applications*, 6th Edition. Pearson, 2021.
3. G. B. Arfken, H. J. Weber, and F. E. Harris, *Mathematical Methods for Physicists: A Comprehensive Guide*, 7th Edition, Academic Press (Elsevier), 2013, ISBN:978-0-12-384654-9.
4. D. G. Zill and W. S. Wright, *Advanced Engineering Mathematics*, 6th Edition. Jones & Bartlett Learning, 2018.

Free/Open-Source AI Tools /Software:

1. Google Colab
2. Python (NumPy, SymPy, SciPy, Matplotlib)
3. Scilab
4. Sage Math
5. GeoGebra

Course Articulation Matrix:

Course Outcomes	Program Outcomes										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2	2	–	–	–	–	–	–
CO2	3	3	3	2	2	–	–	–	–	–	–
CO3	3	3	3	2	2	–	–	–	–	–	–
CO4	3	3	3	3	2	–	–	–	–	–	–
CO5	3	3	3	3	2	–	–	–	–	–	–

Correlation Levels: 3 - High; 2 - Moderate; and 1- Low

I B. Tech I Semester – Common to all Branches

L	T	P	C
3	0	2	4

26ACS01: COMPUTATIONAL THINKING AND PROBLEM SOLVING WITH PYTHON

Course Outcomes (COs)

After successful completion of the course, the student will be able to:

CO1 : Apply computational thinking techniques and Python programming fundamentals to design and implement basic programs using algorithms, flowcharts, pseudocode, variables, data types and operators.

CO2 : Apply decision-making and looping constructs to develop Python programs for solving conditional and iterative computational problems.

CO3 : Analyze problem requirements and apply Python string-handling techniques and core data structures such as lists, tuples and sets to develop data-processing solutions.

CO4 : Analyze computational problems and develop modular, reusable Python solutions using dictionaries, user-defined functions, recursion and lambda functions.

CO5 : Develop Python programs using modules, file handling and exception-handling mechanisms, apply basic object-oriented programming concepts to solve practical problems.

UNIT – I: COMPUTATIONAL THINKING AND PROGRAMMING BASICS

10 Hours

Computational Thinking: Characteristics, problem-solving strategies, steps in problem solving; algorithms - definition and properties; flowcharts - symbols and construction; pseudocode - writing and conversion; abstraction, decomposition, pattern recognition, algorithm efficiency basics.

Introduction to Python: Installation and execution environment; variables, identifiers, keywords; Data types, Type conversion; Input and Output statements; Expressions and Operators - Arithmetic, Relational, Logical, Assignment; Operator precedence.

AI Integration: Use Flowgorithm to design and simulate algorithms (e.g., largest of 3 numbers); Draw flowcharts using Lucidchart; Use Algorithm Visualizer to trace algorithm execution; Use ChatGPT to convert pseudocode to Python and verify output.

UNIT – II: DECISION MAKING AND LOOPING 8 Hours

Control Statements: Boolean expressions; if, if-else, if-elif-else, nested if; Conditional expressions (ternary operator). Looping statements: while loop, for loop, iteration techniques, nested loops, infinite loops; loop control - break, continue, pass; else with loops.

Problem Solving: Prime number check, Pattern programs using nested loops, menu-driven programs using control statements.

AI Integration: Use Python Tutor to visualize if-else and loop execution step by step; Use Thonny to debug loop-based programs, Use ChatGPT to generate test cases for control flow programs.

UNIT – III: STRINGS AND DATA STRUCTURES 9 Hours

Strings: Representation, indexing, slicing, operations, built-in functions and methods.

Lists: Creation, indexing and slicing, operations, functions, methods, nested lists.

Tuples: Creation, operations, packing and unpacking.

Sets: creation, set operations — union, intersection, difference; frozen sets.

AI Integration: Use NLTK and TextBlob to perform text analysis (sentiment, word frequency) on strings; Use Pandas and NumPy for list and array operations; Use ChatGPT to explain string slicing and list comprehensions with concrete examples.

UNIT – IV: FUNCTIONS AND PROBLEM SOLVING 9 Hours

Dictionaries: Creation, operations, methods; dictionary-based applications. Functions: Built-in and user-defined functions; function definition and calling; arguments - positional, keyword, default, variable-length; scope of variables (local and global).

Recursion: Recursive functions - factorial, Fibonacci; lambda functions (anonymous functions); Applications of functions in problem solving.

AI Integration: Use Jupyter Notebook for interactive function experiments; Use SymPy for symbolic mathematical expressions; Use PyCharm for modular program development.

UNIT – V: FILE HANDLING, EXCEPTION HANDLING AND OOP 9 Hours

Modules and Packages: Creating and importing modules; Standard library modules.

File handling: Opening, reading, writing, and closing files; file modes; working with text files; processing CSV and Excel files.

Exception handling: Types of errors (syntax, runtime, logical); try, except, finally blocks; raising exceptions. Introduction to OOP: classes, objects, attributes, methods, constructors, self keyword; basic applications of OOP in Python.

AI Integration: Use Pandas and OpenPyXL to read and write CSV/Excel files; Use Pytest for testing Exception handling; Use ChatGPT to explain OOP class design with real-world examples; Use Django for a simple mini web application as a capstone project.

LIST OF EXPERIMENTS:

Experiment 1: Select three daily-life problems; apply decomposition and abstraction; write algorithms, draw flowcharts, and convert them into pseudocode.

Experiment 2: Design flowcharts for largest of three numbers, student grade calculation, and electricity-bill generation; simulate execution and verify outputs.

Experiment 3: Install the Python environment; write programs using variables, data types, input/output, and operators; solve simple interest, area, and unit-conversion problems.

Experiment 4: Develop Python programs for largest among three numbers, grade calculation, leap-year checking, and a menu-driven calculator.

Experiment 5: Implement programs for prime-number generation, Armstrong numbers, factorial computation, and multiplication tables.

Experiment 6: Generate pyramid, diamond, and character patterns.

Experiment 7: Develop programs for string reversal, palindrome checking, frequency counting, and word-occurrence analysis.

Experiment 8: Perform sorting, searching, list merging, and matrix operations using nested lists.

Experiment 9: Implement tuple packing/unpacking, student-record storage, and set union, intersection, and difference operations.

Experiment 10: Implement positional, keyword, default, and variable-length arguments.

Experiment 11: Develop recursive solutions for Factorial, Fibonacci, GCD, and Tower of Hanoi.

Experiment 12: Implement a student database, dictionary operations, dictionary sorting, and lambda-based computations.

Experiment 13: Develop programs to create and read text files, count words, and process CSV data.

Experiment 14: Implement try-except, finally, and custom exceptions; apply exception handling to division operations, file access, and input validation.

Experiment 15: Develop a Python application using classes, objects, constructors, and methods.

TEXT BOOKS

1. P. Spronck, *Computational Thinking with Python*, Version 1.0.0, 2023.
2. R. Thareja, *Python Programming Using Problem-Solving Approach*, 3rd Edition. New Delhi, India: Oxford University Press, 2025.

REFERENCE BOOKS

1. A. B. Downey, *Think Python: How to Think Like a Computer Scientist,* 3rd Edition, O'Reilly Media, 2024.
2. E. Matthes, *Python Crash Course: A Hands-On, Project-Based Introduction to Programming*, 3rd Edition, No Starch Press, 2023.
3. J. VanderPlas, *Python Data Science Handbook: Essential Tools for Working with Data*, 2nd Edition, O'Reilly Media, 2023.

ONLINE RESOURCES

1. Algorithm & Flowchart: Flowgorithm – flowgorithm.org; Lucidchart lucidchart.com Algorithm Visualizer – algorithm-visualizer.org

2. Programming & Debugging: Python Tutor – pythontutor.com; Thonny – thonny.org Replit – replit.com
3. Data & Scientific: Pandas – pandas.pydata.org; NumPy – numpy.org; SymPy – sympy.org
4. NLP & Text: NLTK – nltk.org; TextBlob – textblob.readthedocs.io
5. Development: Jupyter Notebook – jupyter.org; PyCharm – jetbrains.com/pycharm OpenPyXL – openpyxl.readthedocs.io
6. NPTEL – Problem Solving through Programming in C / Python – nptel.ac.in

Course Articulation Matrix:

Course Outcomes	Program Outcomes											Program Specific Outcomes	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	3	1	3	–	–	–	1	–	–	3	2
CO2	3	3	3	1	3	–	–	–	1	–	–	3	2
CO3	3	3	3	2	3	–	–	–	2	–	–	3	2
CO4	3	3	3	2	3	–	–	1	2	–	–	3	3
CO5	3	3	3	2	3	–	–	2	2	–	–	3	3

Correlation Levels: 3- High; 2- Moderate; and 1- Low

I B. Tech I Semester - Common to All Branches

L	T	P	C
1	0	4	3

26AAI01:AI TOOLS AND APPLICATIONS

Course Outcomes (COs)

After successful completion of the course, students will be able to:

CO1 : Explain the fundamental concepts of artificial intelligence and generative AI, classify and compare AI tools across text, image, audio, video and code applications.

CO2 : Apply prompt-engineering techniques and AI productivity tools to create, refine and evaluate documents, presentations and study materials using appropriate source-grounded workflows.

CO3 : Use AI-assisted research, note-taking and no-code data-analysis tools to retrieve, verify, organize, analyze and interpret information and datasets using appropriate evidence-verification practices.

CO4 : Create and evaluate AI-assisted visual, diagrammatic, audio and video content, identify and validate relevant AI applications for engineering disciplines using appropriate technical or authoritative sources.

CO5 : Evaluate AI applications for bias, privacy, intellectual property, ethical and governance risks, design and present an AI-assisted mini-project integrating appropriate tools and responsible-AI practices.

UNIT – I: UNDERSTANDING AI AND AI FOR DOCUMENTS AND PRESENTATIONS

15 Hours

AI Definition: AI problems, foundation of AI and AI brief history (Turing Test → ChatGPT → generative AI era); how LLMs work (training data, next-token prediction, temperature). AI ecosystem: ChatGPT, Claude, Gemini, Microsoft Copilot- strengths and differences.

AI for presentations: Gamma.app - full slide deck from a paragraph prompt; Microsoft Copilot in PowerPoint - slide generation, design suggestions, speaker notes; Canva AI - Magic Design, text-to-presentation, brand kits; Beautiful.ai -smart slide layouts; comparing AI-generated vs. manually created presentations.

AI for document writing: Microsoft Copilot in Word - draft, rewrite, summarize; Gemini in Google Docs - writing assistance and inline Q&A; Notion AI - meeting notes, project plans, knowledge base; Claude Projects - organizing documents and having a persistent AI assistant over a set of files.

AI writing assistance: Grammarly - grammar, tone, clarity, plagiarism; ChatGPT and Claude for lab reports, proposals, and letters; academic integrity - AI as an assistant not an author; disclosure norms; identifying AI-generated content using GPTZero.

AI Integration: Use Gamma.app to generate a presentation on an engineering topic in under

2 minutes; use Copilot in Word or Gemini in Docs to draft and improve a report; compare ChatGPT, Claude, Gemini, and Microsoft Copilot on the same writing task; use Grammarly to review your own writing.

UNIT – II: AI FOR RESEARCH, NOTE-TAKING AND SPREADSHEETS

15 Hours

NotebookLM (Google): uploading lecture notes, textbooks, PDFs, and research papers as sources; asking questions grounded in uploaded documents; generating study guides, timelines, briefing documents, and audio overviews (podcast-style summaries) from sources; source-grounded answers that cite the exact passage - eliminating hallucination risk.

AI for research: Perplexity AI (real-time web search with citations), Gemini Deep Research (multi-step research reports), Elicit and Consensus (academic paper search with evidence summaries); verifying AI-generated citations on Google Scholar; why AI fabricates references and how to detect it.

Prompt engineering: zero-shot, one-shot, few-shot, chain-of-thought, role-based prompting; iterative refinement; negative prompting; building effective prompts for summarization, comparison, explanation, and data extraction; using ChatGPT custom instructions and Claude Projects for persistent context.

AI for spreadsheets and data analysis: Microsoft Copilot in Excel - VLOOKUP, IF, COUNTIF, pivot tables from natural language; Gemini in Google Sheets - formula generation and chart creation; ChatGPT Advanced Data Analysis and Julius AI - uploading CSV files, computing statistics, generating charts, interpreting patterns - no coding required.

AI Integration: Upload your own lecture notes to NotebookLM and generate a study guide and audio overview; use Perplexity AI and Gemini Deep Research to research a topic; verify citations on Google Scholar; use Wolfram Alpha or Symbolab to solve and verify equations from your current semester subjects; use Copilot in Excel or ChatGPT ADA to analyse a real dataset.

UNIT – III: AI FOR VISUAL DESIGN, DIAGRAMS, AND MEDIA 15 Hours

AI for image generation: DALL-E 3 (via ChatGPT), Canva AI (text-to-image, Magic Design, background remover), Adobe Firefly, Ideogram - writing effective image prompts (style, subject, composition, colour); iterative prompt refinement for posters, diagrams, and mockups.

AI for diagrams and visual thinking: Napkin.ai - converting text or concepts into visual diagrams, charts, and infographics automatically; Whimsical AI - AI-generated mind maps, flowcharts, and wireframes from a description; Miro AI - collaborative whiteboard with AI summarization and diagramming.

AI for audio and video: ElevenLabs - text-to-speech and voice cloning for presentations and voiceovers; Suno and Udio - AI music generation for project videos; Runway and Pika — generating short video clips from text prompts; HeyGen - AI avatars for video presentations; current limitations and appropriate use cases.

AI content detection and critical evaluation: GPTZero, Originality.ai for AI-text detection; Google Lens and reverse image search for AI-generated images; fact-checking workflows - cross-referencing primary sources; deepfakes and synthetic media - recognition, risks, and responsible creation.

AI Integration: Use Napkin.ai to convert a paragraph of text into a diagram; use Whimsical AI to generate a mind map of a course topic; use DALL-E 3 or Canva AI to create an event poster through 3 prompt iterations; use ElevenLabs to create a voiceover; test GPTZero on 5 samples and analyse confidence scores.

UNIT– IV: AI ACROSS ENGINEERING DISCIPLINES 15 Hours

AI in Mechanical Engineering: generative design in CAD (Autodesk Fusion 360 - AI-optimized structures for weight and load); predictive maintenance using sensor data and ML; AI-assisted FEA simulation; manufacturing process optimization; digital twins.

AI in Civil and Electrical Engineering: structural health monitoring using ML; smart grid load forecasting and energy management; AI for circuit simulation and PCB layout optimization (Altium AI); VLSI design automation - floor planning and routing; power systems fault detection.

AI in Electronics and Communication: AI-assisted signal processing and filtering; speech recognition for embedded systems; computer vision for industrial quality inspection; AI in antenna design and spectrum management; MATLAB AI and Deep Learning Toolbox.

AI in Computer Science and allied branches: GitHub Copilot for code generation, explanation, and review; AI-powered testing (Testim, Selenium AI); AI in cybersecurity - anomaly detection and intrusion detection; no-code/low-code platforms (Bubble, Glide, Webflow AI); AI for data science (Jupyter AI, DataRobot).

AI Integration Each student identifies 5 real AI tools in their own engineering branch using Claude or ChatGPT and verifies each using official product pages or research publications; present findings in a 3-minute class presentation with a structured comparison table.

UNIT– V: RESPONSIBLE AI, ETHICS 15 Hours

AI bias and fairness: sources of bias (data, model, deployment); real documented cases - Amazon hiring algorithm (2018), COMPAS recidivism (2016), facial recognition disparities (NIST 2019); fairness metrics; misinformation, deepfakes, and synthetic media risks.

Privacy and intellectual property: what happens to data entered into public AI tools; PII risks; data retention policies of ChatGPT (OpenAI), Claude (Anthropic), and Gemini (Google); who owns AI-generated content; copyright status globally; academic plagiarism with AI; personal data safety protocol.

AI governance frameworks: EU AI Act risk tiers (unacceptable, high, limited, minimal); UNESCO AI Ethics Recommendation; India's NITI Aayog Responsible AI Guidelines; India Digital Personal Data Protection Act 2023 (DPDP Act); environmental cost of AI - energy and carbon footprint of LLM training and inference.

AI Integration: Read and compare data retention policies of ChatGPT, Claude, and Gemini; classify 5 daily-use AI systems using EU AI Act risk tiers; use GPTZero and Google Lens to detect AI-generated content; design and present the Capstone Mini-Project integrating tools from all five units.

LIST OF EXPERIMENTS

Sl. No.	Unit	Title of the Experiment
1	Unit I	ChatGPT & Claude - Understanding LLMs, Temperature and Hallucination; Unit I: Understanding AI and AI for Documents
2		Gamma.app - AI-Generated Presentation from a Paragraph Prompt; Unit I: AI for Presentations
3		Microsoft Copilot in Word - Draft, Rewrite, and Summarize; Unit I: AI for Document Writing
4		Grammarly - Grammar, Tone, Clarity, and Plagiarism Detection; Unit I: AI Writing Assistance
5		GPTZero - AI Text Detection: Test 5 Samples and Analyze Scores; Unit I: Academic Integrity & AI Detection
6		Google Gemini - Multimodal Prompting with Image and Text Input
7	Unit II	NotebookLM - Upload Lecture Notes and Generate Study Guide + Audio Overview; Unit II: NotebookLM
8		Perplexity AI - Real-Time Research with Verified Citations; Unit II: AI for Research
9		Elicit & Consensus - Academic Paper Search and Evidence Synthesis; Unit II: AI for Research
10		Prompt Engineering - Zero-Shot to Chain-of-Thought on Engineering Tasks; Unit II: Prompt Engineering
11		Copilot in Excel / Julius AI - Natural Language Data Analysis of a CSV Dataset; Unit II: AI for Spreadsheets
12		Zotero + SciSpace - Reference Management and AI Paper Q&A
13	Unit III	DALL-E 3 / Canva AI - Event Poster Through 3 Iterative Prompt Refinements; Unit III: AI for Image Generation
14		Napkin.ai & Whimsical AI - Concept Diagram and Course Mind Map; Unit III: AI for Diagrams
15		Miro AI - Collaborative System Architecture Brainstorm; Unit III: AI for Diagrams
16		Eleven Labs - Create a 60-Second Technical Project Voiceover; Unit III: AI for Audio
17		Google Lens & GPTZero - Detect AI-Generated Text and Images; Unit III: AI Content Detection
18		Text-to-Video - Create a 30-Second Technical Explainer with Canva AI / Runway
19	Unit IV	GitHub Copilot - Code Generation, Explanation and Review; Unit IV: AI in Computer Science
20		ChatGPT ADA - Full Data Science Pipeline on a Real Dataset; Unit IV: AI for Data Science
21		Bubble.io or Glide - Build a No-Code AI-Powered App Prototype; Unit IV: AI in CS - No-Code Platforms

22		Wolfram Alpha & Symbolab - Solve and Verify Engineering Equations; Unit IV: AI Math Verification
23		Branch AI Tool Research - Identify and Verify 5 Real Branch - Specific AI Tools; Unit IV: AI Tools Integration
24		Teachable Machine - Train and Deploy a No-Code Image Classifier
25	Unit V	AI Bias Audit - Analyze Compas Fairness Metrics; Unit V: AI Bias and Fairness
26		Privacy Audit - Compare Data Retention Policies of ChatGPT, Claude, Gemini; Unit V: Privacy and AI
27		EU AI Act Classification - Classify 5 Daily-Use AI Systems; Unit V: AI Governance
28		AI Hallucination & Fact-Checking Audit - A Responsible Verification Workflow
29		Capstone Deliverable - Integrated Multi-Tool Engineering Solution; Unit V: Capstone Project
30		Capstone Presentation - 5-Minute Project Demo with Responsible AI Disclosure; Unit V: Capstone Project

TEXT BOOKS:

1. S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th Edition. Pearson, 2022.
2. H. Motwani, *Generative AI 360°: Practical Guide to ChatGPT, Midjourney & AI Tools to Boost Productivity & Creativity: For Professionals,* 1st Edition ZebraLearn, 2025

REFERENCE BOOKS:

1. E. Mollick, *Co-Intelligence: Living and Working with AI*. Penguin, 1st Edition, 2024
2. Chip Huyen, *Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications*, 1st Edition, O'Reilly Media, 2022.
3. Reid Blackman, *Ethical Machines: Your Concise Guide to Totally Unbiased, Transparent, and Respectful AI*, Harvard Business Review Press, 2022
4. F. Chollet, *Deep Learning with Python*, 2nd Edition, Manning Publications, 2021

Course articulation matrix:

Course Outcomes	Program Outcomes											Program Specific Outcomes	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	1	–	–	3	–	–	–	1	–	–	3	2
CO2	3	2	2	1	3	–	–	–	3	–	–	2	3
CO3	3	3	1	3	3	–	–	–	2	–	–	3	3
CO4	3	2	3	2	3	–	–	1	2	–	–	3	3
CO5	3	2	3	2	3	–	–	2	3	–	–	3	3

Correlation Levels: 3- High; 2- Moderate; and 1- Low

I B. Tech I Semester – Common to CE & ME

L	T	P	C
0	0	3	1.5

26AME01: ENGINEERING WORKSHOP

Course Outcomes (COs)

After successful completion of the course, students will be able to:

CO1 : Identify commonly used workshop tools, equipment and materials, apply appropriate safety practices in different workshop trades.

CO2 : Perform basic woodworking, sheet-metal working, fitting and electrical wiring operations to fabricate simple joints, components and electrical connections.

CO3 : Prepare green-sand moulds and perform basic arc-welding and gas-welding operations to produce simple engineering joints and components.

CO4 : Perform basic lathe operations, including facing, turning, step turning, chamfering and drilling, to manufacture simple cylindrical components.

CO5 : Apply the basic principles of additive manufacturing to produce simple mechanical components using a 3D printing machine.

LIST OF EXPERIMENTS

Demonstration: Safety practices and precautions to be observed in workshop.

1. Familiarity with different types of woods and tools used in wood working and make following joints.
   a) Half– Lap joint b) Mortise and Tenon joint
2. Familiarity with different types of tools used in sheet metal working, Developments of following sheet metal job from GI sheets.
   a) Tapered tray b) Conical funnel
3. Familiarity with different types of tools used in fitting and do the following fitting exercises.
   a) V-fit b) Dovetail fit
4. Familiarity with different types of basic electrical circuits and make the following connections.
   a) Parallel and series b) Two-way switch
5. Demonstration and practice on Moulding tools and processes, Preparation of Green Sand Moulds for given Patterns.
6. Demonstration and practice on Arc Welding and Gas welding. Preparation of Lap joint and Butt joint.
7. Basic Machine Shop: Demonstration and practice on lathe Machine – Turning, Facing, Step Turning, Chamfering and Drilling.

8. Introduction to Additive Manufacturing: Demonstration and practice on 3D Printing Machine. Preparation of mechanical components using 3D Printing Machine.

TEXT BOOKS

1. Felix W, *Basic Workshop Technology: Manufacturing Process*, Independently Published, 1st Edition, 2019.
2. B. S. Raghuwanshi, *A Course in Workshop Technology*, Vols. I & II, Dhanpat Rai & Co., 10th Edition, 2023.

REFERENCE BOOKS

1. S. K. Hajra Choudhury et al., *Elements of Workshop Technology*, Vol. I, 17th Edition, 2023, Media Promoters and Publishers, Mumbai.
2. H. S. Bawa, *Workshop Practice*, Tata McGraw-Hill, 2nd Edition 2023.
3. P. M. Soni and P. A. Upadhyay, *Wiring Estimating, Costing and Contracting*, Atul Prakashan, 1st Edition, 2026.

Course Articulation Matrix:

Course Outcomes	Program Outcomes											Program Specific Outcomes	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	2	1	3	-	-	2	2	-	-	2	3
CO2	3	3	3	2	3	-	-	3	2	-	-	2	3
CO3	3	3	3	3	3	-	-	3	2	-	-	2	3
CO4	3	3	3	3	3	-	-	3	2	-	-	2	3
CO5	3	3	3	3	3	-	-	3	3	-	-	2	3

Correlation Levels: 3- High; 2- Moderate; and 1- Low

I B.Tech. I Sem- Common to All Branches

L	T	P	C
0	0	1	0.5

26AHS08 - HEALTH AND WELLNESS, YOGA AND SPORTS

Course Outcomes (COs)

After the completion of the course the student will be able to

- CO1 : Explain and apply basic principles of health, nutrition, balanced diet and physical fitness in maintaining personal wellness.
- CO2 : Demonstrate appropriate yoga practices, including Asanas, Pranayama, Kriya, Mudra, Bandha and Dhyana for physical and mental well-being.
- CO3 : Demonstrate physical fitness and sports skills through participation in selected individual and team sporting activities and fitness exercises.
- CO4 : Demonstrate discipline, sportsmanship, emotional balance and a positive attitude while participating in sports, yoga and wellness activities.
- CO5 : Demonstrate teamwork and leadership skills through participation in sports, health-awareness and wellness activities.
- CO6 : Adopt healthy, responsible and sustainable lifestyle practices that contribute to personal and social well-being.

UNIT – I:

Concept of health and fitness, Nutrition and Balanced diet, basic concept of immunity Relationship between diet and fitness, Globalization and its impact on health, Body Mass Index (BMI) of all age groups.

ACTIVITIES:

i) Organizing health awareness programmes in community
ii) Preparation of health profile
iii) Preparation of chart for balance diet for all age groups

UNIT – II:

Concept of yoga, need for and importance of yoga, origin and history of yoga in Indian context, classification of yoga, Physiological effects of Asanas- Pranayama and meditation, stress management and yoga, Mental health and yoga practice.

ACTIVITIES:

Yoga practices – Asana, Kriya, Mudra, Bandha, Dhyana, Surya Namaskar

UNIT – III:

Concept of Sports and fitness, importance, fitness components, history of sports, Ancient and Modern Olympics, Asian games and Commonwealth games.

ACTIVITIES:

i) Participation in one major game and one individual sport viz., Athletics, Volleyball, Basketball, Handball, Football, Badminton, Kabaddi, Kho-kho, Table tennis, Cricket etc.
ii) Practicing general and specific warm up, aerobics
iii) Practicing cardiorespiratory fitness, treadmill, run test, 9 min walk, skipping and running.

GENERAL GUIDELINES:

1 . Institutes must assign slots in the Timetable for the activities of Health/Sports/Yoga.
2. Institutes must provide field/facility and offer the minimum of five choices of as many as Games/Sports.
3. Institutes are required to provide sports instructor / yoga teacher to mentor the students.

EVALUATION GUIDELINES:

- Evaluated for a total of 100 marks.
- A student can select 6 activities of his/her choice with a minimum of 01 activity per unit. Each activity shall be evaluated by the concerned teacher for 15 marks, totalling to 90 marks.
- A student shall be evaluated by the concerned teacher for 10 marks by conducting viva voce on the subject.

TEXT BOOKS:

1. Ministry of Youth Affairs and Sports, Government of India, NSS Training Manual. New Delhi, India: MYAS Publication, 2022.
2. Directorate General NCC, National Cadet Corps - Training Syllabus & Manual, New Delhi, India: DG NCC, 2021.

REFERENCE BOOKS:

1. S. K. Lal, Community Development & Social Service. New Delhi, India: Regal Publications, 2019.
2. S. Modh, Disaster Management – A Holistic Approach. New Delhi, India: Macmillan India, 2020.
3. S. McCurley and R. Lynch, Volunteer Management. Energize Publications, 2018.
4. R. Baden-Powell, Scouting for Boys, Centenary ed. Oxford, U.K.: Oxford University Press, 2008.

Course Articulation Matrix:

Course Outcomes	Program Outcomes										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	1	–	–	–	–	2	1	1	–	–	3
CO2	1	–	–	–	–	2	–	1	1	–	3
CO3	1	–	–	–	–	1	–	2	3	1	2
CO4	1	–	–	–	–	2	–	3	3	2	2
CO5	1	–	–	–	–	2	–	2	3	3	2
CO6	1	–	–	–	–	3	2	2	2	1	3

Correlation Levels: 3- High; 2- Moderate; and 1- Low

I B.Tech. II Sem- Common to CE and ME

L	T	P	C
3	0	2	4

26AHS04: ENGINEERING PHYSICS

Course Outcomes (COs):

After successful completion of the course, students will be able to:

CO1 : Apply the principles of wave optics to analyze interference, diffraction and polarization phenomena, interpret experimental results for engineering applications.

CO2 : Apply crystallographic and material-characterization principles to determine and interpret crystal structure, lattice parameters and related structural properties of engineering materials using X-ray diffraction techniques.

CO3 : Demonstrate the principles of laser physics and optical fibres through appropriate experiments, determine optical parameters such as laser wavelength, numerical aperture and acceptance angle, evaluate their engineering applications in sensing and communication.

CO4 : Explain and apply the principles of oscillations and acoustics to interpret vibration, resonance, damping, sound intensity, reverberation and sound absorption, evaluate the basic requirements of acoustically suitable engineering spaces.

CO5 : Apply the principles of ultrasonics and basic non-destructive testing methods to characterize engineering materials, assess quality, detect surface defects in engineering components.

UNIT – I: WAVE OPTICS 10 Hours

Interference: Interference of light - Conditions for sustained interference - Interference due to thin films (division of amplitude) - Newton's rings - Determination of wavelength and refractive index of a liquid.

Diffraction**:** Introduction to diffraction, Fresnel and Fraunhofer diffraction. Fraunhofer diffraction due to single slit and double slits (Qualitative), Diffraction grating - Dispersive power and Resolving power of grating. Applications: Measurement of distances, angles and surface characterization.

Polarization**:** Polarization of light - production of Polarized light - reflection, refraction and double refraction - Nicol's prism – applications of polarized light (Photoelasticity, Optical surveying instruments, Remote sensing, Non-destructive testing (NDT) and Anti-glare optical systems).

AI Integration: AI-Assisted Flatness Detection Using Interference Fringes, AI-Based Mineral Analysis and Degradation Detection in Masonry and Concrete Using Diffraction. AI integration holographic interferometry for micro-deformation analysis. Python-based fringe pattern analysis using image processing.

UNIT – II: CRYSTALLOGRAPHY AND MATERIAL CHARACTERIZATION

7 Hours

Crystallography**:** Space lattice, Basis, Unit Cell and lattice parameters - Crystal systems, Bravais Lattices - Coordination number - Packing fraction of SC, BCC & FCC - Miller indices - Separation between successive (h k l) planes.

X-ray diffraction**:** Bragg's law - X-ray Diffractometer – Crystal structure determination by Powder diffraction method. Application: Checking the quality of construction materials.

AI Integration: AI-driven XRD for Real-time corrosion assessment in reinforced concrete and identifying microcracks in ageing infrastructure. AI-assisted XRD pattern analysis for phase identification and crystal structure determination. ML-based classification of diffraction patterns using scikit-learn.

UNIT – III: LASERS AND OPTICAL FIBRES 9 Hours

Lasers**:** Principle of lasing action – Characteristics - Einstein theory of stimulated emission - Population inversion - Pumping Schemes - Three and four level lasers - Components of laser - Construction and working of He-Ne Laser and Nd-YAG Laser, engineering applications of lasers.

Optical Fibres: Introduction to optical fibres - Total Internal Reflection - critical angle - construction of optical fibres - Acceptance angle, Acceptance cone - Numerical Aperture - Classification of fibres based on modes of propagation and choice of materials - fabrication of optical fibre by double crucible technique, applications of optical fibres - sensors (temperature, displacement, liquid level detector).

AI Integration: Laser control - AI-based feedback control - Automatic adjustment of laser power and wavelength; AI Integration in Optical Fibers: Machine-Learning-Based Prediction of Fiber Attenuation and Dispersion.

UNIT – IV: OSCILLATIONS AND ACOUSTICS 10 Hours

Oscillations**:** Introduction to oscillations, Series and parallel combinations of springs in Suspension Systems, Stiffness factor and its physical significance, vibration isolation. Theory of damped oscillations (qualitative), Forced oscillations (qualitative) and Resonance; Sharpness of resonance and Quality factor (Q).

Acoustics**:** Introduction, classification of sound, sound intensity level (decibel), reverberation, reverberation time, Sabine's formula (qualitative), absorption coefficient and its determination, Factors affecting acoustics and their remedies, basic requirements of an acoustically good hall.

AI Integration: AI-Assisted Adaptive Suspension System for EVs, AI-Assisted Strategies for noise mitigation in Green Buildings. Use of Acoustic Simulation AI for auditorium design.

UNIT – V: ULTRASONICS AND NON-DESTRUCTIVE TESTING 9 Hours

Ultrasonics**:** Introduction to ultrasonic waves, production of ultrasonic waves by magnetostriction method, piezoelectric method, detection of ultrasonics (qualitative), applications of ultrasonics for engineering materials and structures.

Non-destructive testing**:** Introduction, role of NDT in Engineering structures. Basic testing methods-Visual inspection - 3D recording the image of an object and Liquid penetrant testing: basic principle for detecting surface defects in steel.

AI Integration: AI-Based Noise Reduction and Signal Enhancement, Ultrasonic Thickness and Corrosion Monitoring AI-Based Vibration AI-Powered Defect Detection in Structural Steel, Weld Joints. Automatic defect detection from images and signals.

LIST OF EXPERIMENTS:

1. Determination of refractive index of a given material using Brewster's law.
2. Determination of radius of curvature of a plano convex lens using Newton's rings method.
3. Determination of wavelengths of various colours of mercury spectrum using diffraction grating in the normal incidence method.
4. Determination of Miller indices of different crystal planes using a crystal model or a suitable simulation and to understand the orientation of planes in a crystal lattice.
5. Determination of the wavelength of a LASER using a diffraction grating.
6. Estimation of the numerical aperture and acceptance angle of an optical fiber.
7. Verify the oscillations of a mass under different combinations of springs (Loaded Springs).
8. Sonometer: Verification of the three laws of stretched strings.
9. Determination of spring constant using Hook's law.
10. Determination of rigidity Modulus of a material of a wire by a dynamic method - Torsional Pendulum.
11. Verify the generation of ultrasonic waves using the magnetostriction method.
12. Determination of ultrasonic velocity of a liquid. (Using Acoustic grating).
13. Detection and Evaluation of Surface Defects in Engineering Materials by Visual Inspection Method.

TEXT BOOKS:

1. M. N. Avadhanulu, and T. V. S. Arun Murthy, *A Textbook of Engineering Physics*, 11th Edition, S. Chand Publishing, 2026.
2. B. K. Pandey and S. Chaturvedi, *Engineering Physics*, 2nd Edition, Cengage Learning, 2023.
3. S. Balasubramanian, S. Ranganathan and M.N. Srinivasan, *A Text book of Practical Physics*, 2nd Edition, Sultan Chand Publishers, New Delhi, 2017.

REFERENCE BOOKS:

1. V. Rajendran, *Engineering Physics*, 2nd Edition, Mc Graw Hill, 2026.
2. R.K. Shukla and Anchal Srivastava, *Experimental Physics*, 1st Edition, New age international publishers, 2017.
3. P.K. Palaniswamy, *Engineering Physics*, 4th Edition, Scitech Publications, 2016.
4. R. Jayaraman, V. Umadevi, S, Maruthamuthu, D. Sarvanakumar, *Engineering physics Laboratory Manual*,1st Edition, Pearson, 2013.

ONLINE RESOURCES

1. https://vlab.amrita.edu/index.php?sub=1&brch=194&sim=354&cnt=1
2. https://home.iitk.ac.in/~anjankg/virtual-lab/phy-ltl/?utm_source=chatgpt.com

Course Articulation Matrix:

Course Outcomes	Program Outcomes										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	3	3	-	-	2	2	-	-
CO2	3	3	3	3	3	-	-	2	2	-	-
CO3	3	3	3	3	3	-	-	2	3	-	-
CO4	3	3	3	3	3	-	-	2	2	-	-
CO5	3	3	3	3	3	-	-	2	2	-	-

Correlation Levels: 3- High; 2- Moderate; and 1- Low

I B. Tech II Semester - Common to CE & ME

L	T	P	C
3	0	2	4

26ACE01: ENGINEERING MECHANICS

Course Outcomes (COs)

After successful completion of the course, students will be able to:

CO1 : Apply the fundamental principles of mechanics, force systems, friction, equilibrium and virtual work to solve engineering problems involving particles, rigid bodies, trusses and frames.

CO2 : Determine the centroid, centre of gravity, area moment of inertia and mass moment of inertia of regular and composite sections and bodies, apply these concepts to engineering problems.

CO3 : Analyze the kinematics and kinetics of particles using equations of motion, work-energy, impulse-momentum and impact principles to solve engineering problems.

CO4 : Analyze stresses, strains and deformation in simple and composite members using elastic constants, factor of safety, Mohr's circle and strain-energy principles.

CO5 : Apply mechanics principles to model, solve and interpret engineering problems involving force systems, structural members and particle motion using analytical and computational tools.

UNIT – I: INTRODUCTION, FORCE SYSTEMS & FRICTION 9 Hours

Basic Concepts of Mechanics: scalar and vector quantities, Newton's laws, SI units. Coplanar concurrent forces: resultant by parallelogram law, triangle law, polygon law. Moment of a force: Varignon's theorem. Couples and resultants of force systems. Lami's theorem. Spatial force systems: vector representation, dot and cross products. Friction: limiting friction, Coulomb's laws of dry friction, coefficient of friction, angle of friction, cone of static friction, impending motion problems, wedge friction, screw jack.

AI Integration: Introduction to AI in Engineering: concept of data, machine learning (ML), and simulation. Using Python (NumPy) to solve force equilibrium problems computationally. Introduction to MATLAB /Scilab for vector force calculations. Visualizing force systems using Matplotlib /Python.

UNIT – II: EQUILIBRIUM OF FORCE SYSTEMS & VIRTUAL WORK 9 Hours

Equilibrium conditions for coplanar systems: free body diagrams, equations of equilibrium ($\Sigma Fx=0$, $\Sigma Fy=0$, $\Sigma M=0$). Graphical methods: triangle and polygon of forces. Equilibrium of spatial systems. Analysis of plane trusses: method of joints, method of sections. Analysis of frames. Principle of virtual work: statement, applications to simple mechanisms and trusses.

AI Integration: Using Python/OpenSees (open source) for automated truss analysis. AI-based structural analysis: introduction to how FEM software uses matrix methods internally. Computer simulation of mechanism motion using Python. Visual debugging of FBDs using interactive tools.

UNIT – III: CENTROID, CENTRE OF GRAVITY & MOMENT OF INERTIA 9 Hours

Centroid of lines, areas, and volumes from first principles. Centroid of standard shapes: rectangle, triangle, circle, semicircle, quarter circle. Centroid of composite figures using tabular method. Centre of gravity of 3D bodies: prismatic bodies, cones, hemispheres, composite bodies. Pappus theorems and their applications. Area moment of inertia: polar moment, transfer theorem (parallel axis theorem). MOI of composite sections. Product of inertia. Mass moment of inertia for simple and composite bodies.

AI Integration: Using spreadsheets (LibreOffice Calc) and Python Pandas for tabular calculations. Introduction to how CAD software (AutoCAD/FreeCAD) computes section properties. Application in AI-driven structural design optimization: cross-section selection.

UNIT – IV: KINEMATICS & KINETICS OF PARTICLES 9 Hours

Kinematics: rectilinear motion – displacement, velocity, acceleration relations; curvilinear motion – Cartesian and polar coordinates; projectile motion. Kinetics: Newton's second law; D'Alembert's principle; work-energy method: kinetic energy, work done, power, efficiency; impulse-momentum method: linear and angular; impact: direct and oblique central impact, coefficient of restitution.

AI Integration: AI in Autonomous Vehicles: how kinematic equations underpin self-driving car algorithms. Python simulation of projectile motion with drag (real-world). ML-based trajectory prediction: introduction to LSTM for motion prediction. Robot path planning using kinematic constraints. Using Python (matplotlib.animation) for interactive motion visualization.

UNIT – V: SIMPLE STRESSES, STRAINS & ELASTIC CONSTANTS 9 Hours

Elasticity and plasticity; types of stresses (tensile, compressive, shear) and strains (linear, lateral, shear, volumetric); Hooke's law; Poisson's ratio; elastic constants: E, G, K and their relationships; factor of safety; bars of varying cross-section; elongation under self-weight; bars in series and parallel; temperature stresses; stresses in composite bars. Introduction to principal stresses and strains: principal planes, Mohr's circle for 2D stress state. Strain Energy-Resilience–Gradual, sudden, impact and shock loadings – simple applications.

AI Integration: ML for material property prediction: using regression models to predict Young's modulus and Poisson's ratio from material composition data. Python-based stress-strain curve fitting using SciPy. Introduction to data-driven material models in structural analysis. Overview of physics-informed neural networks (PINNs) for material mechanics (conceptual).

LIST OF EXPERIMENTS

Experiment-1: Verification of axial forces in the members of a truss
Experiment-2: Experimental Proof of Lami's Theorem
Experiment-3: Verification of polygon law of forces
Experiment-4: Verification of triangle law & parallelogram law of forces
Experiment-5: Verification of newton's laws of motion
Experiment-6: Determination of Center of Gravity of different shaped Plane Lamina.
Experiment-7: Determination of coefficient of Static and Rolling Friction.
Experiment-8: Verification of Law of Moment using Rotation Disc Apparatus and Bell Crank Lever
Experiment-9: Determination of coefficient of friction between two surfaces
Experiment-10: Verification of motion parameters using conservation of energy.
Experiment-11: Safety Practices in Construction Industry
Experiment-12: Study of Alternative Materials in Modern construction.
Experiment-13: To Study the Application of Lami's Theorem for construction site.
Experiment-14: Field-visit to construction site for quality testing and assessment.

TEXT BOOKS:

1. S. Timoshenko, D. H. Young, J. V. Rao, and S. Pati, *Engineering Mechanics*: In SI Units, 5th Edition, New Delhi, India: McGraw Hill Education, 2017.
2. S. S. Bhavikatti, *A Textbook of Engineering Mechanics*, 4th Edition, New Delhi, India: New Age International Publishers, 2018.
3. S. Timoshenko and J. M. Gere, *Mechanics of Materials*. 2nd Edition New Delhi, India: CBS Publishers, 2011.
4. R. K. Rajput, *A Textbook of Strength of Materials: Mechanics of Solids*, 8th Edition, New Delhi, India: S. Chand Publishing, 2025.
5. B. S. Basavarajaiah and P. Mahadevappa, *Strength of Materials*, 3rd Edition, Hyderabad, India: Universities Press, 2010.

REFERENCE BOOKS:

1. R. C. Hibbeler, *Engineering Mechanics: Statics & Dynamics*, 15th Edition, Hoboken, NJ, USA: Pearson, 2022.
2. J. L. Meriam, L. G. Kraige, and J. N. Bolton, *Engineering Mechanics: Statics*, 9th Edition, Hoboken, NJ, USA: John Wiley & Sons, 2020.
3. F. P. Beer, E. R. Johnston, D. F. Mazurek, P. J. Cornwell, and B. Self, V*ector Mechanics for Engineers: Statics and Dynamics*, 13th Edition, New Delhi, India: McGraw Hill, 2026.
4. B. Bhattacharya, Introduction to Statics and Dynamics, 2nd Edition, Oxford, U.K.: Oxford University Press, 2014.
5. R. V. Giles, J. B. Evett, and C. Liu, Schaum's Outline of Fluid Mechanics and Hydraulics, 4th Edition, New York, NY, USA: McGraw Hill, 2013.

ONLINE RESOURCES

NPTEL – Engineering Mechanics, IIT Madras – https://nptel.ac.in/courses/112106286

Course Articulation Matrix:

Course Outcomes	Program Outcomes											Program Specific Outcomes	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	2	–	2	-	-	2	2	-	-	2	2
CO2	3	3	2	–	2	-	-	2	2	-	-	2	2
CO3	3	3	2	2	2	-	-	2	2	-	-	2	2
CO4	3	3	3	2	2	-	-	2	2	-	-	2	2
CO5	3	3	2	2	3	-	-	2	2	-	-	2	2

Correlation Levels: 3- High; 2- Moderate; and 1- Low

I B. Tech II Semester- Common to All Branches

L	T	P	C
3	0	0	3

26AHS07: DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS

Course Outcomes

After Successful completion of the course, the student will be able to:

CO1 : Solve first-order ordinary differential equations using appropriate analytical methods and apply them to model engineering processes such as cooling, growth / decay and electrical circuits.

CO2 : Analyze higher-order linear ordinary differential equations using appropriate solution methods and apply them to model dynamic systems such as electrical circuits, oscillators and mass-spring-dashpot systems.

CO3 : Formulate and solve first- and second-order partial differential equations using appropriate analytical methods, classify and interpret PDE models such as the heat, wave and Laplace equations.

CO4 : Compute and analyze the gradient, directional derivative, divergence and curl of Scalar and vector fields, apply vector identities and field concepts to engineering problems.

CO5 : Evaluate line and surface integrals and apply Green's, Stokes' and Divergence Theorems to compute work, circulation, flux, area and volume in engineering applications.

UNIT-I: FIRST-ORDER ORDINARY DIFFERENTIAL EQUATIONS 9 Hours

Classification of ODEs by order, degree, and linearity; linear first-order ODEs and integrating factor method; Bernoulli's equation; exact equations and integrating factors; orthogonal trajectories; Newton's law of cooling and natural growth/decay models; electrical circuit models (RC and RL circuits); engineering applications of first-order ODEs.

AI Integration: Python SymPyd solve for symbolic ODE solution and verification; direction field (slope field) visualization using Matplotlib; AI-assisted debugging of integrating factor derivations; using ChatGPT/Copilot to explain solution families and physical interpretations.

UNIT-II: HIGHER-ORDER LINEAR DIFFERENTIAL EQUATIONS 9 Hours

Superposition principle; Wronskian and linear independence; complementary function for constant-coefficient ODEs (distinct real, repeated, and complex roots); particular integral by operator method; variation of parameters; simultaneous linear ODEs; Cauchy's (Euler) equation; Legendre's equation; applications to LCR circuits, simple harmonic motion,

damped and forced oscillations; mass-spring-dashpot systems**.**

AI Integration: Python SciPy signal. 1sim for simulation of under damped, critically damped, and over damped responses; resonance amplitude curve generation; AI-assisted Wronskian computation and verification using SymPy; using AI tools to explain transient vs. steady-state response; resonance and Q-factor interpretation.

UNIT - III: PARTIAL DIFFERENTIAL EQUATIONS 9 Hours

Introduction and classification of PDEs; formation by elimination of arbitrary constants and functions; first-order linear PDEs and Lagrange's method; homogeneous linear PDEs with constant coefficients (CF and PI); standard forms of nonlinear first-order PDEs; second-order PDE applications - wave equation, heat equation, and Laplace's equation; classification as hyperbolic, parabolic, and elliptic.

AI Integration: Python SymPy for PDE formation and symbolic verification; 3D surface plot visualization of PDE solution families using Matplotlib; SciPy finite-difference simulation of the 1D heat equation with animation; AI-assisted PDE classification using the discriminant; prompting AI to classify mixed-derivative PDEs and student verification of AI reasoning.

UNIT - IV: VECTOR DIFFERENTIATION 9 Hours

Scalar and vector fields; del (∇) operator; gradient and directional derivative; unit - normal to surfaces; angle between surfaces; divergence and physical interpretation; curl and physical interpretation; vector identities; conservative fields and scalar potentials; Laplacian; engineering applications in electromagnetic and fluid mechanics contexts.

AI Integration: Python NumPy and Matplotlib for 2D and 3D gradient, divergence, and curl field visualization (quiver plots, level-surface overlays); SymPy for symbolic verification of vector identities; using AI tools to interpret Maxwell’s equations in terms of divergence and curl operators; AI-assisted explanation of irrotational and solenoidal field classification.

UNIT - V: VECTOR INTEGRATION AND INTEGRAL THEOREM 9 Hours

Line integrals over scalar and vector fields; work done and path independence; surface integrals and flux; Green's theorem in the plane and its applications; Stokes' theorem; Divergence theorem (Gauss's theorem); applications to area and volume computation; engineering applications in electromagnetism, fluid mechanics, and mass conservation.

AI Integration: Python SciPy integrate for numerical computation of line and surface integrals; 3D flux visualization with colour-coded normal flux using Matplotlib; numerical verification of Green’s and Stokes’ theorems; AI-assisted physical interpretation of the Divergence theorem in terms of source density; prompting AI to explain connections to Gauss’s law and student evaluation of AI explanations.

TEXT BOOKS:

1. E. Kreyszig, *Advanced Engineering Mathematics*, 11th Edition, International Adaptation. Hoboken, NJ, USA: Wiley, 2025.
2. B. S. Grewal, *Higher Engineering Mathematics*, 45th Edition, New Delhi, India: Khanna Publishers, 2024.
3. J. R. Hass, C. E. Heil, M. D. Weir, and P. Bogacki, *Thomas' Calculus*, 15th Edition, Pearson, 2022.

REFERENCE BOOKS:

1. R. Haberman and P. Choboter, *Applied Partial Differential Equations*, 6th Edition, Boston, MA, USA: Pearson, 2025.
2. D. G. Zill and W. S. Wright, *Advanced Engineering Mathematics*, 6th Edition, Burlington, MA, USA: Jones & Bartlett Learning, 2018.
3. I. N. Sneddon, *Elements of Partial Differential Equations*. Mineola, NY, USA: Dover Publications, 2006.
4. G. B. Arfken, H. J. Weber, and F. E. Harris, *Mathematical Methods for Physicists: A Comprehensive Guide*, 7th Edition, Academic Press (Elsevier), 2013, ISBN:978-0-12-384654-9.

Free/Open-Source AI Tools /Software:

1. Google Colab
2. Python (NumPy, SymPy, SciPy, Matplotlib)
3. Scilab
4. Sage Math
5. GeoGebra

Course Articulation Matrix:

Course Outcomes	Program Outcomes										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2	2	–	–	–	–	–	–
CO2	3	3	3	3	2	–	–	–	–	–	–
CO3	3	3	3	3	2	–	–	–	–	–	–
CO4	3	3	3	3	2	–	–	–	–	–	–
CO5	3	3	3	3	2	–	–	–	–	–	–

Correlation Levels: 3- High; 2-Moderate; and 1- Low

I B. Tech II Semester – Common to all Branches

L	T	P	C
3	0	2	4

26ACS02: DATA STRUCTURES USING PYTHON

Course Outcomes (COs)

After successful completion of the course, the student will be able to:

CO1 : Analyze abstract data types and data structures using time and space complexity measures, implement and compare basic searching and sorting algorithms.

CO2 : Implement stacks and queues using appropriate representations and apply them to expression evaluation, parenthesis matching, scheduling and other computational problems.

CO3 : Implement and manipulate singly, doubly and circular linked lists, apply them to practical problems such as polynomial processing, browser history and symbol tables.

CO4 : Analyze and implement binary trees, binary search trees and heaps, apply tree traversal, searching and heap operations, including heap sort, to computational problems.

CO5 : Represent and traverse graphs using adjacency matrices/lists, BFS and DFS, implement hash tables using appropriate collision-resolution techniques.

UNIT – I: INTRODUCTION, ALGORITHM ANALYSIS AND LINEAR STRUCTURES 10 Hours

Introduction to data structures, Abstract Data Types (ADTs); Algorithm analysis - time and space complexity; Big-O, Big-Omega, Big-Theta notations.

Lists in Python: operations on lists; static vs. dynamic memory allocation.

Searching techniques: Linear Search, Binary Search, Interpolation Search -analysis and comparison.

Sorting techniques: Bubble Sort, Selection Sort, Insertion Sort - algorithms, trace, and complexity analysis.

AI Integration: Use VisuAlgo and Algorithm Visualizer to animate sorting and searching; Use ChatGPT to explain Big-O complexity with concrete examples; Use Python Tutor to trace memory representation of data structures.

UNIT – II: STACKS AND QUEUES 8 Hours

Stacks: LIFO principle, push, pop, peek; Implementation using Python lists; applications - Expression evaluation (infix to postfix, postfix evaluation), parenthesis checking.

Queues: FIFO principle, enqueue, dequeue; types - Circular Queue, Priority Queue, Deque; implementation using Python lists and linked structures. Applications of queues: BFS simulation, scheduling, real-world analogies.

AI Integration: Use VisuAlgo to animate stack and queue operations; Use ChatGPT to generate edge cases (overflow, underflow, circular behaviour); Use GitHub Copilot to implement expression evaluation.

UNIT – III: LINKED LISTS 10 Hours

Singly Linked List: Node structure, insertion at head/tail/position, deletion by value/position, traversal, reversal.

Doubly Linked List: Forward and backward traversal, insertion, deletion; Circular Linked List: traversal and operations.

Applications: Polynomial representation and addition, browser history simulation (doubly linked list), symbol table implementation.

AI Integration: Use Python Tutor to trace pointer operations and node memory; Use VisuAlgo to visualize insertion and deletion step by step.

UNIT - IV: TREES AND HEAPS 9 Hours

Trees: Terminology, properties; Binary Trees - Representation, traversals (Inorder, Preorder, Postorder, level order).

Binary Search Trees (BST): Insertion, deletion, search; applications - autocomplete, range queries, dictionary lookup.

Heaps: Min-Heap and Max-Heap, array representation, heapify, Build-Heap; Heap Sort - algorithm and complexity.

AI Integration: Use VisuAlgo and Algorithm Visualizer to animate tree traversals and BST operations; Use GitHub Copilot to implement heap operations in Python.

UNIT – V: GRAPHS AND HASHING 9 Hours

Graphs: Terminology, types (directed, undirected, weighted); representation - adjacency matrix and adjacency list.

Graph traversals: BFS (queue-based) and DFS (stack/recursion-based); Applications - shortest path, cycle detection, social networks.

Hashing: Hash functions, properties of a good hash function; collision resolution - chaining (separate chaining), open addressing (linear probing, quadratic probing, double hashing); load factor.

AI Integration: Use NetworkX in Python and Gephi to build and traverse graphs; Use VisuAlgo to animate BFS and DFS; Use GitHub Copilot to implement hash tables with collision resolution.

LIST OF EXPERIMENTS

Experiment-1: Implement insertion, deletion, searching, and slicing; compare static and dynamic memory allocation using Python lists.

Experiment-2: Implement Linear Search, Binary Search, and Interpolation Search; compare

execution tie and number of comparisons.

Experiment-3: Implement Bubble Sort, Selection Sort, and Insertion Sort; visualize execution and compare performance on different input datasets.

Experiment-4: Implement stack operations using Python lists and solve parenthesis-matching problems.

Experiment-5: Implement infix-to-postfix conversion and postfix evaluation.

Experiment-6: Implement Queue, Circular Queue, Priority Queue, and Deque.

Experiment-7: Simulate BFS traversal and scheduling applications using queue structures.

Experiment-8: Implement insertion, deletion, traversal, and reversal operations for Single Linked List.

Experiment-9: Implement forward and backward traversal, insertion, and deletion for Double and Circular Linked List.

Experiment-10: Develop polynomial addition using linked lists and simulate browser history navigation.

Experiment-11: Implement Inorder, Preorder, Postorder, and level-order traversals for bianry Tree.

Experiment-12: Implement BST insertion, deletion, and search; implement Min Heap, Max Heap, and Heap Sort.

Experiment-13: Represent graphs using adjacency matrix and adjacency list; implement BFS and DFS.

TEXT BOOKS

1. S. K. Vasudevan, A. S. Nagarajan, and K. Nanmaran, *Data Structures Using Python,* 1st Edition, New Delhi, India: Wiley India, 2021.
2. Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser , *Data Structures and Algorithms in Python*, 1st Edition. New Delhi, India: Wiley India, 2021.

REFERENCE BOOKS

1. R. Thareja, *Python Programming: Using Problem-Solving Approach*, 3rd Edition, Oxford University Press, 2025.
2. E. Matthes, *Python Crash Course: A Hands-On, Project-Based Introduction to Programming,* 3rd Edition San Francisco, CA, USA: No Starch Press, 2022.
3. B. N. Miller, D. L. Ranum, and R. Yasinovskyy, *Problem Solving with Algorithms and Data Structures Using Python,* 3rd Edition. Franklin, Beedle & Associates, 2023.
4. T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, *Introduction to Algorithms*, 4th Edition, MIT Press, 2022.

ONLINE RESOURCES

1. NPTEL – Data Structures and Algorithms – Nptel.ac.in
2. VisuAlgo – visualgo.net
3. Python Tutor – pythontutor.com
4. NetworkX – networkx.org
5. GeeksforGeeks – geeksforgeeks.org
6. Algorithm Visualizer – algorithm-visualizer.org

Course Articulation Matrix:

Course Outcomes	Program Outcomes											Program Specific Outcomes	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	3	2	3	–	–	–	1	–	–	3	2
CO2	3	3	3	2	3	–	–	1	1	–	–	3	3
CO3	3	3	3	2	3	–	–	1	1	–	–	3	3
CO4	3	3	3	2	3	–	–	1	1	–	–	3	3
CO5	3	3	3	2	3	–	–	1	2	–	–	3	3

Correlation Levels: 3-High; 2-Moderate; and1-Low

I B.Tech. II Sem - ME

L	T	P	C
3	0	0	3

26AEE02: BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

Course Outcomes (COs):

After successful completion of the course a students will be able to

CO1: Remember the fundamental laws, electrical quantities, operating principles of electrical machines, measuring instruments, and Energy systems.

CO2: Understand AC/DC circuit behavior, operation of machines and measuring instruments, electricity billing concepts, power generation methods, and electrical safety practices.

CO3: Apply mathematical methods, electrical engineering principles, and AI-assisted tools to solve problems related to circuits, machines, and energy systems.

CO4: Analyze the characteristics of PN junction diode, Zener diode, and BJT configurations to evaluate their behaviour and select appropriate devices for basic electronic applications.

CO5: Apply knowledge of rectifiers, filters, amplifiers, and instrumentation systems to explain circuit operation, logic gates, Boolean and simulate waveform behaviour using Python tools.

PART A: BASIC ELECTRICAL ENGINEERING

UNIT- I: DC & AC CIRCUITS 9 Hours

DC Circuits: Electrical circuit elements (R, L and C), Ohm's Law and its limitations, KCL & KVL, series, parallel, series-parallel circuits, Super Position theorem, Simple Numerical problems.

AC Circuits: A.C. Fundamentals: Equation of AC Voltage and current, waveform, time period, frequency, amplitude, phase, phase difference, average value, RMS value, form factor, peak factor, Voltage and current relationship with phasor diagrams in R, L, and C circuits, Concept of Impedance, Analysis of R-L, R-C, R-L-C Series circuits, Active power, reactive power and apparent power, Concept of power factor (Simple Numerical problems).

AI Integration: Introduction to AI-assisted Electrical Circuit Analysis, Intelligent Waveform and Phasor Diagram Analysis, Machine Learning Applications for Voltage and Current Prediction, AI-based Power Factor Monitoring and Correction, AI-enabled Circuit Simulation Tools and Applications

Free/Open-Source AI Tools: Python (NumPy, SciPy, Matplotlib) on Google Colab; Ngspice; Jupyter Notebook

UNIT – II: MACHINES AND MEASURING INSTRUMENTS 9 Hours

Machines: Construction, principle and operation of (i) DC Motor, (ii) DC Generator, (iii) Single Phase Transformer, (iv) Three Phase Induction Motor and (v) Alternator, Applications of electrical machines.

Measuring Instruments: Construction and working principle of Permanent Magnet Moving Coil (PMMC), Moving Iron (MI) Instruments and Wheat Stone bridge.

AI Integration: Introduction to AI in Electrical Machines, AI-based Predictive Maintenance of Motors and Transformers, Smart Sensor-based Monitoring of Electrical Parameters, AI-based Error Detection in Measuring Instruments, Intelligent Digital Measuring Instruments

Free/Open-Source AI Tools: Python (NumPy, SciPy, Matplotlib) on Google Colab; Ngspice; Jupyter Notebook

UNIT - III: ENERGY RESOURCES, ELECTRICITY BILL & SAFETY MEASURES 9 Hours

Energy Resources: Conventional and non-conventional energy resources; Layout and operation of various Power Generation systems: Hydel, Nuclear, Solar & Wind power generation.

Equipment Safety Measures: Working principle of Fuse and Miniature Circuit Breaker (MCB), merits and demerits. Personal safety measures: Electric Shock, Earthing and its types, Safety Precautions to avoid shock.

AI Integration: Introduction to Smart Energy Systems and AI Applications, Smart Electrical, Safety Monitoring Systems.

Free/Open-Source AI Tools: Python (NumPy, SciPy, Matplotlib) on Google Colab; Ngspice; Jupyter Notebook

TEXT BOOKS

1. D. C. Kulshreshtha, *Basic Electrical Engineering*, 2nd edition. New Delhi, India: Tata Mc Graw-Hill, 2019.
2. P. V. Gupta, M. L. Soni, U. S. Bhatnagar, and A. Chakrabarti, *Power System Engineering*. 2nd edition, New Delhi, India: Dhanpat Rai & Co., 2010.
3. R. Prasad, *Fundamentals of Electrical Engineering*, 3rd edition. New Delhi, India: Prentice Hall of India, 2014.

REFERENCE BOOKS

1. D. P. Kothari and I. J. Nagrath, *Basic Electrical Engineering*, 4th Edition. New Delhi, India: McGraw-Hill, 2019.
2. V. K. Mehta and S. Mehta, *Principles of Power Systems*. 4th Edition, New Delhi, India: S. Chand Technical Publishers, 2022.
3. T. K. Nagsarkar and M. S. Sukhija, *Basic Electrical Engineering*. 3rd Edition, Oxford, U.K.: Oxford University Press, 2019.
4. S. K. Bhattacharya, *Basic Electrical and Electronics Engineering*, 4th Edition, New Delhi, India: Pearson, 2017.

PART B: BASIC ELECTRONICS ENGINEERING:

UNIT – I: SEMICONDUCTOR DEVICES 9 Hours

Introduction — Evolution of electronics: vacuum tubes to nano electronics. Characteristics of PN Junction Diode. Zener Effect — Zener diode and its characteristics. Bipolar Junction Transistor — CB, CE, CC configurations and characteristics. Elementary treatment of small signal CE amplifier.

AI Integration:

1. Use AI/LLM tools to explain the construction of a PN junction diode, and the Zener effect; ask it to describe the V-I characteristics of a PN junction and Zener diode with sketches, and compare their operating regions in a short table.
2. Use a provided Python template in Google Colab to plot the ideal V-I characteristic curve of a PN junction diode using the diode equation; vary the saturation current and thermal voltage and observe how the curve shape changes, then superimpose the Zener diode characteristic to compare forward and reverse breakdown behaviour.
3. Use AI/LLM tool to explain CB, CE, and CC BJT configurations — input/output characteristics, current gain, and typical applications; prepare a comparison table covering input impedance, output impedance, current gain, voltage gain, and phase relationship, and ask the LLM to describe the small signal CE amplifier operation in simple terms.

UNIT – II: BASIC ELECTRONIC CIRCUITS AND INSTRUMENTATION

9 Hours

Block diagram of a DC power supply, working of a full wave bridge rectifier, capacitor filter (no analysis), working of simple Zener voltage regulator. Amplifiers: block diagram of Public Address system, circuit diagram and working of common emitter (RC coupled) amplifier with frequency response, concept of voltage divider biasing. Electronic Instrumentation: block diagram of an electronic instrumentation system.

AI Integration:

1. Use AI/LLM tool to explain the block diagram of a DC power supply and the working of a full wave bridge rectifier; ask it to describe the role of a capacitor filter and a Zener voltage regulator in simple terms, and prepare a labelled block diagram description with the function of each stage (transformer, rectifier, filter, regulator).
2. Use a provided Python template in Google Colab to simulate a full wave bridge rectifier output: generate a sinusoidal input signal, apply ideal diode rectification, and plot the input and rectified output waveforms; then add a smoothing capacitor and observe how the ripple voltage changes with different capacitor values.
3. Use AI/LLM tool to explain the working of a common emitter RC coupled amplifier - the role of coupling capacitors, bypass capacitor, and voltage divider bias network and sketch the gain-vs-frequency curve, then prepare a short note on the block diagram of an electronic instrumentation system.

UNIT – III: DIGITAL ELECTRONICS 9 Hours

Overview of number systems. Logic gates including universal gates. BCD codes, Excess-3 code, Gray code, Hamming code. Boolean algebra — basic theorems and properties. Truth tables and functionality of logic gates: NOT, OR, AND, NOR,
NAND, XOR, and XNOR. Simple combinational circuits: half adder and full adder. Introduction to sequential circuits — flip-flops, registers, and counters (elementary treatment only).

AI Integration:

1. Use an AI/LLM tool to explain the binary, octal, and hexadecimal number systems and conversions between them; ask it to describe BCD, Excess-3, Gray, and Hamming codes with examples, and solve 2–3 numerical problems on code conversion (e.g., decimal to BCD, binary to Gray code); verify the results manually.
2. Use a provided Python template in Google Colab to generate truth tables for all seven logic gates (NOT, OR, AND, NOR, NAND, XOR, XNOR) for 2-variable and 3-variable inputs; verify De Morgan's theorems programmatically by comparing both sides of the identities, and implement a half adder and full adder using basic gate functions.
3. Use an AI/LLM tool to explain SR, D, JK, and T flip-flops in simple terms — truth tables, excitation tables, and the concept of clock-triggered operation; ask it to describe the working of a 4-bit shift register and a 3-bit ripple counter with state sequences, and prepare a concise comparison table of flip-flop types.

TEXT BOOKS

1. R. L. Boylestad and L. Nashelsky, *Electronic Devices and Circuit Theory*, 11th edition. Upper Saddle River, NJ, USA: Pearson Education, 2013.
2. T. L. Floyd, *Digital Fundamentals*, 11th edition. Boston, MA, USA: Pearson Education, 2021.

REFERENCE BOOKS

1. A. S. Sedra and K. C. Smith, *Microelectronic Circuits*, 8th edition. New York, NY, USA: Oxford University Press, 2019.
2. J. Millman and C. C. Halkias, *Electronic Devices and Circuits*. 4th edition New Delhi, India: Tata McGraw-Hill, 2015.
3. M. M. Mano and M. D. Ciletti, *Digital Design*, 6th edition. Boston, MA, USA: Pearson Education, 2017.

Course Articulation Matrix:

Course Outcomes	Program Outcomes											Program Specific Outcomes		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	1	1	–	2	-	-	-	-	-	-	3	2	2
CO2	3	2	1	1	2	-	-	-	-	-	-	2	2	2
CO3	3	3	3	2	3	-	-	-	-	-	-	3	2	2
CO4	3	3	3	2	3	-	-	-	-	-	-	2	2	2
CO5	3	3	3	2	3	-	-	-	-	-	-	2	3	3

Correlation levels: 3 – High; 2 – Moderate; 1 – Low.

I B. Tech II Semester – Common to CE, EEE & ME

L	T	P	C
1	0	4	3

26AME02: ENGINEERING GRAPHICS

Course Outcomes (COs)

After successful completion of the course, the student will be able to:

CO1 : Construct engineering curves and scales using appropriate graphical methods and apply standard engineering graphics conventions in technical drawings.

CO2 : Construct and interpret orthographic projections of points, lines and planes in different orientations with respect to the reference planes, determine their true lengths and inclinations.

CO3 : Construct and interpret orthographic projections of regular solids in different positions and orientations with respect to the reference planes.

CO4 : Construct sectional views and determine the true shapes of sections of solids, develop surfaces of common engineering solids using parallel-line and radial-line development methods.

CO5 : Construct and interpret isometric and orthographic views of simple and compound solids, create basic 2D and 3D engineering drawings using AutoCAD.

UNIT – I: INTRODUCTION TO ENGINEERING GRAPHICS, CURVES, AND SCALES

9 Hours

Introduction: Principles of Engineering Graphics, signboards, BIS conventions, and lettering.

Curves: Construction of Polygons. Conic sections: Ellipse, Parabola, and Hyperbola using eccentricity/general methods. Cycloidal curves and Involutes.

Scales: Plain, Diagonal, and Vernier scales.

Hands-on Experience: Manual draft generation of an ellipse and diagonal scale, paired with a software verification loop.

AI Integration Module: AI prompt engineering for generating geometric profiles; algorithmic generation of conic curves using mathematical code blocks; parsing hand-drawn geometric sketches into clean vector shapes using Computer Vision (CV) edge detection (Canny Edge/Hough Transform).

UNIT – II: ORTHOGRAPHIC PROJECTIONS OF POINTS, LINES

9 Hours

Core Syllabus: Principles of orthographic projections, projection of points in all four quadrants. Projection of straight lines inclined to one or both reference planes, traces of lines, determination of true lengths and true inclinations.

Projections of Planes: Regular planes Perpendicular to both reference planes, parallel to one

reference plane and inclined to the other reference plane; plane inclined to both the reference planes.

Hands-On Experience & Application

Project: Construct a transformation script where a user inputting 2D projection endpoints automatically outputs the 3D line representation, true spatial length, and inclination angles without manual geometric tracking.

Application: Calibration setups for stereoscopic robotic arms tracking trajectories across multiple camera planes.

AI Integration Module: Machine learning pipelines for 3D coordinate estimation from 2D views; multi-view structural tracking; training neural models to predict true line length and orientation angles based solely on coordinate projections.

UNIT – III: PROJECTION OF REGULAR SOLIDS 9 Hours

Projection of Regular Solids: Types of solids: Polyhedra and Solids of revolution. Projections of solids in simple positions: Axis perpendicular to horizontal plane, Axis perpendicular to vertical plane and Axis parallel to both the reference planes, Projection of Solids with axis inclined to one reference plane and parallel to another plane.

Hands-On Experience & Application

Project: Set up a classification notebook using scikit-learn or a simple convolutional model to automatically identify the type of solid (e.g., hexagonal prism vs. cone) present within an orthographic engineering layout image.

Application: Automated component sorting inside factory staging bays and conveyor streams.

AI Integration Module: Object recognition models (e.g., YOLO / Mobile Net) trained to identify standard solid geometries; automated Multiview projection generation from 3D CAD meshes using programmatic render layers.

UNIT - IV: SECTIONS OF SOLIDS, DEVELOPMENT OF SURFACES, AND AI GENERATIVE DESIGN 9 Hours

Sections of Solids: Perpendicular and inclined section planes, Sectional views and True shape of section, Sections of solids in simple position only. Development of Surfaces: Methods of Development: Parallel line development and radial line development. Development of a cube, prism, cylinder, pyramid and cone.

AI Integration Module: Generative Adversarial Networks (GANs) for generating flat-pattern configurations; algorithmic optimization of surface layouts to minimize sheet manufacturing scrap; predictive toolpath generation using physics-guided heuristic networks.

Hands-On Experience & Application

Project: Use a nesting optimization script (e.g., using genetic algorithms or standard heuristics) to arrange several custom flat-surface solid developments onto a finite raw sheet grid, actively minimizing material margins and cut lines.

Application: Sheet metal stamping pattern automation in automotive and aerospace chassis manufacturing.

UNIT – V: ISOMETRIC PROJECTIONS, CONVERSIONS, AND 2D-TO-3D DEEPP RECONSTRUCTION 9 Hours

Isometric Projections: Principles of isometric projection, isometric scale, isometric views/projections of planes, simple and compound solids. Conversion of isometric views to orthographic views and vice-versa.

Computer graphics: Creating 2D&3D drawings of objects including PCB and Transformations using Auto CAD (Not for end examination).

Hands-On Experience & Application

Project: Use a pretrained depth estimation or pixel-to-voxel neural model to parse a set of front, top, and side drawings, compiling them directly into an interactive 3D point cloud or boundary surface rendering.

Application: Instantly generating 3D printable mechanical files directly from archival 2D layout drafting logs.

Note: The practice will be carried out by using AutoCAD software.

TEXT BOOKS

1. N. D. Bhatt, *Engineering Drawing*, Charotar Publishing House, 55th Edition, 2025.
2. Dhananjay Jolhe, *Engineering Drawing with an Introduction to AutoCAD*, Tata McGraw Hill, 1st Edition, 2017.

REFERENCE BOOKS

1. Pradeep Jain, A. P. Gautam, and Abhishek Bhargava, *Engineering Graphics and Design*, Khanna Publishing. 2nd Edition, 2024.
2. K. L. Narayana and P. Kannaiah, *Engineering Drawing*, Tata McGraw Hill, 4th Edition, 2021.
3. M. B. Shah and B. C. Rana, *Engineering Drawing*, Pearson Education India, 2nd Edition 2014.

ADDITIONAL LEARNING RESOURCES

1. https://onlinecourses.nptel.ac.in/e-learning/preview/noc21_me128

Course Articulation Matrix:

Course Outcomes	Program Outcomes											Program Specific Outcomes	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	3	1	2	–	–	–	3	–	–	2	3
CO2	3	3	3	2	2	–	–	–	3	–	–	2	3
CO3	3	3	3	2	2	–	–	–	3	–	–	2	3
CO4	3	3	3	3	2	–	–	–	3	–	–	2	3
CO5	3	3	3	3	3	–	–	2	3	–	–	2	3

Correlation Levels: 3- High; 2- Moderate; and 1- Low

I B. Tech II Semester – ME

L	T	P	C
0	0	3	1.5

26AME03: BASIC SIMULATION LAB

Course Outcomes (COs)

After successful completion of the course, the student will be able to:

CO1 : Develop and execute basic MATLAB scripts for engineering data processing, numerical computation and analysis.

CO2 : Generate and interpret appropriate graphical representations of engineering data to analyze trends and present simulation results.

CO3 : Solve basic algebraic equations, numerical integration problems and ordinary differential equations using appropriate computational tools.

CO4 : Use AI-assisted tools to identify and troubleshoot basic MATLAB coding errors and evaluate the correctness of AI-generated computational solutions.

CO5 : Apply computational and simulation tools to model and analyze simple engineering systems and interpret their results.

LIST OF EXPERIMENTS

EXPERIMENT-1: Introduction to MATLAB Programming

Introduction to the MATLAB interface (Command Window, Workspace), basic calculator functions, variables, creating simple vectors/arrays, basic arithmetic operations, and simple 2D line plotting.

Hands-on Experience: Write a simple script to calculate the area and volume of various mechanical components (cylinders, spheres) and plot a basic sine wave.

AI Applications: Code explanation, fixing basic syntax typos, and generating simple code templates.

AI Integration: Use MATLAB AI Chat Playground to ask: "Explain what this basic error message means and how to fix it."

EXPERIMENT-2: Curve Fitting – Straight Line Fit

Concept of a line of best fit (y = mx + c), basic data importing from Excel, and introduction to the interactive Curve Fitting App.

Hands-on Experience: Import data from a standard Hooke's Law experiment (Force vs. Extension) and find the stiffness (k) of a spring using the visual app interface.

AI Applications: Spotting outliers or bad data points in an experimental dataset.

AI Integration: Use the interactive Curve Fitting App to automatically generate the linear code without writing it from scratch.

EXPERIMENT-3: Curve Fitting – Polynomial Fit

Moving beyond straight lines, understanding non-linear trends, basic concept of a quadratic equation curve, and evaluating fit quality visually.

Hands-on Experience: Fit a smooth curve to a car's displacement-time data to visually observe acceleration trends.

AI Applications: Simple pattern matching and trend prediction based on historical data.

AI Integration: Use the Regression Learner App to explore how a machine learning model can "train" on points to guess a curve.

EXPERIMENT-4: Numerical Integration and Differentiation – Trapezoidal Method

Understanding integration as the simple "area under a curve", dividing an area into basic strips, using the built-in trapz command.

Hands-on Experience: Estimate the total work done by a constant or changing force acting over a distance by calculating the area under a Force-Displacement graph.

AI Applications: Approximating geometry dimensions from digital sensor data streams.

AI Integration: Use generative AI tools to write the logic for a basic for-loop that adds up the areas of separate shapes.

EXPERIMENT-5: Numerical Integration and Differentiation – Simpson Method

Introduction to Simpson's rule as a smarter way to find areas using curved tops instead of flat trapezoids, comparing accuracy.

Hands-on Experience: Find the cross-sectional area of an irregular mechanical bracket or a river bed profile using standard measurement intervals.

AI Applications: Smart shape recognition and automated area estimation in basic design software.

AI Integration: Compare results from the Trapezoidal script and Simpson script using an AI assistant to analyze which method is more accurate.

EXPERIMENT-6: Linear and Nonlinear Equations

Solving a system of basic linear equations (like finding two unknown forces), finding roots of an equation graphically, using simple commands like roots or zero.

Hands-on Experience: Find the precise point where a projectile hits the ground by calculating the zero-crossing root of its trajectory equation.

AI Applications: Finding optimal balance points in manufacturing systems or structural designs.

AI Integration: Use the interactive Optimization Tool wizard to find the minimum weight of a box without manually calculating derivatives.

EXPERIMENT-7: Ordinary Differential Equations & Transforms

Introduction to differential equations as "rates of change" (e.g., speed changing over time), step-by-step calculation using basic Euler logic, using the ode45 app wizard.

Hands-on Experience: Simulate a hot cup of coffee cooling down over time according to Newton's Law of Cooling.

AI Applications: Simple time-series trend forecasting (e.g., predicting when a component will overheat).

AI Integration: Use the Live Editor Task for Solving ODEs to define and solve the cooling rate visually without deep programming.

EXPERIMENT-8: Engineering Applications Analysis (Basic Mechanics Focus)
Simulating fundamental concepts from First-Year Engineering Mechanics: static equilibrium, basic lever arms, and simple velocity profiles.
Hands-on Experience: Plot how the velocity of a falling object changes over time when affected by gravity and simple air resistance.
AI Applications: Identifying unexpected changes or failures in basic mechanical setups.
AI Integration: Use the System Identification App to upload data from a basic pendulum swing and let the tool guess its length and drag.
Video Resources: Engineering Mechanics Animations using MATLAB (Online Lectures).
EXPERIMENT-9: MATLAB Simulink – Introduction to Graphical Modelling
Introduction to the drag-and-drop Simulink interface, working with basic blocks (Sources, Sinks, Gains, Integrators), and connecting signals visually.
Hands-on Experience: Build a visual block diagram model of a basic water tank filling up over time, without writing any lines of code.
AI Applications: An introduction to modern digital twins and graphical system simulations.
AI Integration: Use Simulink Onramp interactive widgets to discover how pre-trained machine learning blocks can predict the output of a system.

TEXT BOOKS

1. Holly Moore, *MATLAB for Engineers*, 6th Edition, 2025.
2. Agam Kumar Tyagi, *MATLAB and SIMULINK for Engineers*. 1st Edition, 2012.
3. Dr. Shailendra Jain, *Modeling and Simulation using MATLAB – Simulink*. 2nd Edition, 2016.

REFERENCE BOOKS

1. Steven C. Chapra and Raymond P. Canale, *Numerical Methods for Engineers*, McGraw Hill Education India. 8th Edition, 2021.
2. John H. Mathews and Kurtis D. Fink, *Numerical Methods Using MATLAB*. 4th Edition, 2015.
3. Won Y. Yang, Wenwu Cao, Tae-Sang Chung, and John Morris, *Applied Numerical Methods Using MATLAB*. 2nd Edition, 2020.

ADDITIONAL LEARNING RESOURCES

1. https://matlabacademy.mathworks.com/details/matlab-onramp/gettingstarted
2. https://matlabacademy.mathworks.com/details/simulink-onramp/simulink
3. https://www.khanacademy.org/math/differential-equations
4. https://matlabacademy.mathworks.com/details/machine-learning-onramp/machinelearning
5. https://matlabacademy.mathworks.com/details/deep-learning-onramp/deeplearning

Course Articulation Matrix:

Course Outcomes	Program Outcomes											Program Specific Outcome	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	2	2	3	-	-	1	2	-	-	2	3
CO2	3	3	2	3	3	-	-	2	3	-	-	2	3
CO3	3	3	3	3	3	-	-	1	2	-	-	2	3
CO4	3	3	2	3	3	-	-	2	3	-	-	2	3
CO5	3	3	3	3	3	-	-	2	3	-	-	2	3

Correlation Levels: 3- High; 2- Moderate; and 1- Low

I B.Tech. II Sem- Common to All Branches

L	T	P	C
0	0	1	0.5

26AHS09 - NSS/NCC /SCOUTS AND GUIDES

Course Outcomes (COs)

After the completion of the course the student will be able to

CO1 : Demonstrate discipline, self-confidence, responsibility and commitment to the values of service and civic responsibility.

CO2 : Identify and analyse basic needs and problems of local communities through observation, surveys and interaction with community stakeholders.

CO3 : Participate in planning and implementing community-service activities addressing identified social, health and environmental concerns.

CO4 : Demonstrate teamwork, leadership, communication and interpersonal skills while organizing and executing community-oriented activities.

CO5 : Promote environmental awareness and sustainable practices through activities related to waste management, recycling, pollution prevention and sustainable living.

CO6 : Demonstrate ethical conduct, empathy and social responsibility towards fellow citizens and disadvantaged sections of society.

UNIT – I: ORIENTATION

General Orientation on NSS/NCC/ Scouts & Guides/Community Service activities, career guidance.

ACTIVITIES:

i) Conducting –ice breaking sessions-expectations from the course-knowing personal talents and skills
ii) Conducting orientations programs for the students –future plans-activities-releasing road map etc.
iii) Displaying success stories-motivational biopics- award winning movies on societal issues etc.
iv) Conducting talent show in singing patriotic songs-paintings- any other contribution.

UNIT – II: NATURE & CARE ACTIVITIES

i) Best out of waste competition.
ii) Poster and signs making competition to spread environmental awareness.
iii) Recycling and environmental pollution article writing competition.
iv) Organising Zero-waste day.
v) Digital Environmental awareness activity via various social media platforms.

vi) Virtual demonstration of different eco-friendly approaches for sustainable living.
vii) Write a summary on any book related to environmental issues.

UNIT – III: COMMUNITY SERVICE ACTIVITIES

i) Conducting One Day Special Camp in a village contacting village-area leaders-Survey in the village, identification of problems- helping them to solve via media-authorities-experts-etc.
ii) Conducting awareness programs on Health-related issues such as General Health, Mental health, Spiritual Health, HIV/AIDS,
iii) Conducting consumer Awareness. Explaining various legal provisions etc.
iv) Women Empowerment Programmes- Sexual Abuse, Adolescent Health and Population Education.
v) Any other programmes in collaboration with local charities, NGOs etc.

TEXT BOOKS:

1. S. K. Mangal and S. Mangal, *Health and Physical Education*. New Delhi, India: PHI Learning Pvt. Ltd., 2021.
2. W. J. Broad, *The Science of Yoga*. New York, NY, USA: Simon & Schuster, 2012.

REFERENCE BOOKS:

1. Nirmalya Kumar Sinha & Surajit Majumder, *A Text Book of National Service Scheme* Vol; I, Vidya Kutir Publication, 2021 (ISBN 978-81-952368-8-6)
2. *Red Book - National Cadet Corps* – Standing Instructions Vol I & II, Directorate General of NCC, Ministry of Defence, New Delhi
3. Davis M. L. and Cornwell D. A., "Introduction to Environmental Engineering", McGraw Hill, New York 4/e 2008
4. Masters G. M., Joseph K. and Nagendran R. "Introduction to Environmental Engineering and Science", Pearson Education, New Delhi. 2/e 2007
5. Ram Ahuja. *Social Problems in India*, Rawat Publications, New Delhi.

GENERAL GUIDELINES:

1. Institutes must assign slots in the Timetable for the activities.
2. Institutes are required to provide instructor to mentor the students.

EVALUATION GUIDELINES:

- Evaluated for a total of 100 marks.
- A student can select 6 activities of his/her choice with a minimum of 01 activity per unit. Each activity shall be evaluated by the concerned teacher for 15 marks, totalling to 90 marks.
- A student shall be evaluated by the concerned teacher for 10 marks by conducting viva voce on the subject.

Course Articulation Matrix:

Course Outcomes	Program Outcomes										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	1	–	–	–	–	2	3	3	2	1	2
CO2	1	3	1	1	–	3	2	2	3	2	2
CO3	1	2	2	1	1	3	3	3	3	2	2
CO4	1	–	1	–	–	2	–	3	3	3	2
CO5	1	2	2	1	1	3	3	2	2	2	3
CO6	1	–	–	–	–	3	3	3	3	2	3

Correlation Levels: 3- High; 2- Moderate; and 1- Low